Is a Description Worth a Thousand Demonstrations? Measuring the Effective Sample Size of Task Descriptions in In-Context Learning

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Abstract

Prompts for in-context learning usually pair demonstrations with a task description, yet theories of in-context learning mostly model the demonstrations alone. This raises a basic question: how many demonstrations is a description worth? We call this number the description's *effective sample size* (ESS) and study it in in-context linear regression, where the Bayes-optimal ESS is known. For the Bayes-optimal predictor, the ESS depends on the description's *specificity*, how much uncertainty about the task it removes, and its *reliability*, the probability that it is relevant. We prove that a reliable description's ESS grows with the task dimension and with its precision, while a description that may be irrelevant is worth a bounded number of demonstrations when none are shown, and trusting it fully can be worse than ignoring it. Meta-trained Transformers come close to the Bayes-optimal ESS and take their trust from training. Pretrained language models answering without reasoning respond to a description's stated specificity far less than the Bayes-optimal predictor, even when nothing else changes. Models that reason first follow it, and two come close to the Bayes-optimal ESS.

1. Introduction

Prompts often tell a model what the task is before, or instead of, showing it. A few-shot prompt for in-context learning (ICL) pairs a natural-language task description with demonstrations (Brown et al., 2020), and deployed models also receive descriptions as system prompts, task instructions, and the instruction files that configure coding agents (Chatlatanagulchai et al., 2025). Yet little is known about how a model uses such a description, and the evidence is mixed. A prompt can be worth hundreds of data points in prompt-based fine-tuning (Le Scao & Rush, 2021), and a written summary can stand in for many demonstrations (Honda et al., 2025). At the same time, models learn as fast from irrelevant or misleading instructions as from instructive ones (Webson & Pavlick, 2022), and they let in-context evidence outweigh explicit instructions (Gupta et al., 2025). Coding agents given repository instruction files do not generally solve more tasks (Gloaguen et al., 2026).

Theories of ICL, in contrast, mostly model the prompt as a sequence of demonstrations (Xie et al., 2022; Garg et al., 2022; Akyürek et al., 2023). Those that add a description show that a learner can use it to predict better (Huang & Ge, 2025), but they do not put a value on it or say what it is worth when it may be wrong. In this work, we measure the value of a task description by the number of demonstrations it can replace, and we ask what determines that number.

We take the Bayesian view of ICL (Xie et al., 2022). The prompt is evidence about a latent task, and a demonstration is one noisy observation of it. A description is a statement about which task generated the demonstrations, so it sharpens the prior; we set aside instruction following. Bayesian statistics measures the information in a prior by its effective sample size (ESS), the number of observations that would carry the same information (Morita et al., 2008). We define the ESS of a description in the same way (Figure 1). We compute it in noisy in-context linear regression with a Gaussian task prior (Garg et al., 2022; Akyürek et al., 2023; Raventós et al., 2023), where Bayes-optimal prediction has a closed form, and then measure it for meta-trained Transformers and for pretrained language models given a stated prior.

Following the Bayesian literature on informative priors, we characterize a description by two properties. The first is its *specificity*, the precision of the prior it gives (Morita et al., 2008; Neuenschwander et al., 2020). "Write this report like an accountant" only loosely indicates the task, while a description that fixes the report's sections, figures, and accounting standard all but states it. The second is its *reliability*, the probability that it is relevant. Reliability sets the weight of the description's prior in a robust mixture that guards against conflict with the data (Schmidli et al., 2014;

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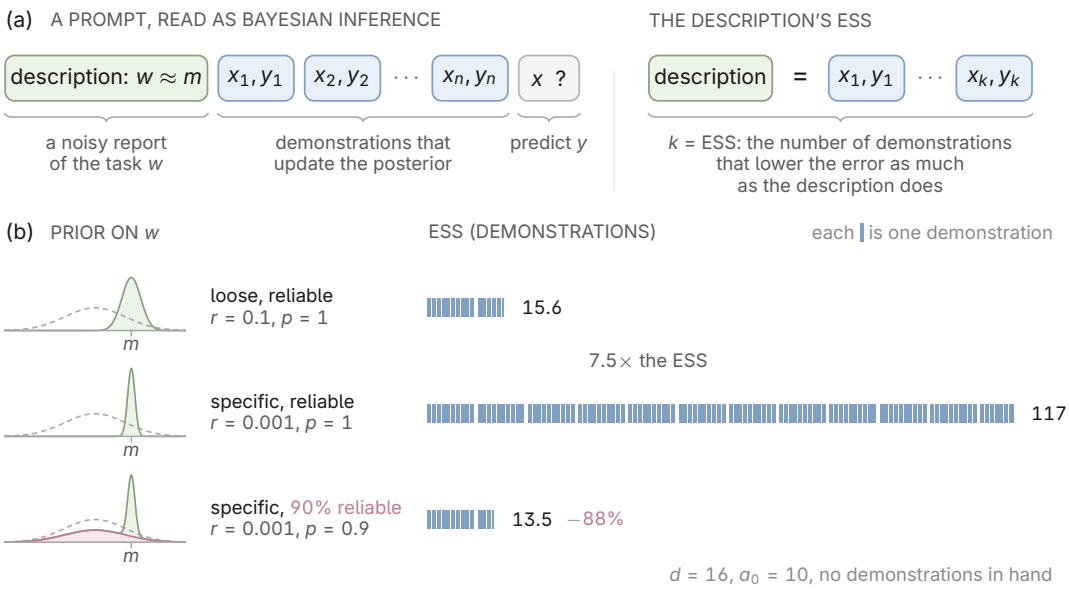


Figure 1. **A description is worth a number of demonstrations, set by its specificity and its reliability.** (a) A prompt read as Bayesian inference: a description states that the task w is near a value m, and the demonstrations update the posterior. The ESS of the description is the number of demonstrations it replaces. (b) The prior on w without the description (dashed) and given three descriptions (solid; sketched, not to scale), and the ESS of each with no demonstrations in hand, at task dimension $d = 16$ and prior signal-to-noise ratio $a_0 = 10$; r is the fraction of the prior variance a relevant description leaves, p the probability that it is relevant, and pink marks the component for an irrelevant description. Specificity raises the ESS (§3); a one-in-ten chance of being irrelevant removes most of that gain (§4).

Evans & Moshonov, 2006). Instructions written for one report can be pasted into a prompt for another, or simply be wrong. Our contributions are as follows.

- **A common unit.** Following data-dependent ESS for priors (Wiesenfarth & Calderazzo, 2020; Reimherr et al., 2021), we define the ESS of a task description as the number of demonstrations that lower a learner's error as much as the description does (Definition 1). The definition applies to any learner, and with squared error it needs only the learner's answers.
- **Theory.** For the Bayes-optimal predictor, a reliable description's ESS grows linearly in the task dimension when the description is loose and in its precision when it is specific (Proposition 1). The dimension term, which the information-based ESS of a prior lacks, fades as demonstrations accumulate. A description of k of the d directions is worth about k without demonstrations, whatever its precision, and $(1 - r)ck/d$ with many (Proposition 2). Using the law of total variance, we show that, without demonstrations, a description that may be irrelevant is worth a bounded number of demonstrations however specific it is, and that once demonstrations are shown, trusting it fully can be worse than ignoring it (Proposition 3, Corollary 4).
- **Experiments.** Meta-trained Transformers come close to the Bayes-optimal ESS, and their trust in a description comes from training (§5). Pretrained language models answering without reasoning respond to a description's stated specificity far less than the Bayes-optimal predictor, even in a control where only the stated specificity changes, and they do not adjust to its stated reliability as it does. A description worth 36 demonstrations to the Bayes-optimal predictor is worth 1 to 8 to them. Models that reason first follow the stated specificity, and Claude Haiku 5.5 and GPT-5.6-Luna come close to the Bayes-optimal ESS (§6).

2. Problem Setup

Tasks and demonstrations. Each task corresponds to a latent weight vector $w \in \mathbb{R}^d$. A demonstration is an input $x \sim \mathcal{N}(0, I_d)$ with a label $y \sim \mathcal{N}(w^\top x, 1)$, so labels are measured in units of their noise. The task has the baseline prior $w \sim \mathcal{N}(0, a_0 I_d)$, and a_0 is the prior signal-to-noise ratio. Unless specified otherwise, the numerical results of §§3–5 use $a_0 = 10$.

Descriptions. A description is a vector $m \in \mathbb{R}^d$ whose relation to the task is governed by two parameters, r and p. A latent indicator $Z \sim \text{Bernoulli}(p)$, independent of m, says whether the description is *relevant* ($Z = 1$) or *irrelevant* ($Z = 0$):

$$
\begin{aligned}
m &\sim \mathcal{N}(0, (1 - r)\, a_0 I_d) \\
w \mid m, Z = 1 &\sim \mathcal{N}(m, r a_0 I_d) \\
w \mid m, Z = 0 &\sim \mathcal{N}(0, a_0 I_d)
\end{aligned}
$$

Intuitively, a relevant description says that the task lies near m, within a variance ra_0 in every direction, while an irrelevant one is unrelated to the task. In both cases w marginally has the baseline prior, and marginalizing over Z gives the robust mixture prior $p\,\mathcal{N}(m, ra_0 I_d) + (1-p)\,\mathcal{N}(0, a_0 I_d)$ (Schmidli et al., 2014). The parameter $r \in (0,1)$ is the fraction of the prior variance that a relevant description leaves. It sets the description's **specificity**: the smaller r, the more specific the description, from one that only loosely indicates the task (r near 1) to one that all but states it (r near 0). The parameter p is the description's **reliability**, and we call a description with $p = 1$ *reliable*. We write $c = 1/(ra_0)$ for the precision of a relevant description's prior, in units of demonstrations, and call a description *loose* when c is small next to d and *specific* when c exceeds d. For example, at $d = 16$ and $a_0 = 10$, $r = 0.1$ ($c = 1$) is loose and $r = 0.001$ ($c = 100$) is specific.

Error and ESS. A predictor sees a prompt D (the description, n demonstrations, and a query x) and predicts the query's label. As Garg et al. (2022) do, we score its prediction $\hat{y}$ by its error, the squared distance from the noiseless label $w^\top x$:

$$R = \mathbb{E}\big[(\hat{y} - w^\top x)^2\big], \tag{1}$$

the expected squared error less the noise variance. Let $R^{\mathrm{desc}}(n)$ be a predictor's error with the description and n demonstrations, and $R^{\mathrm{demo}}(n)$ its error with n demonstrations alone. In §§3–4 the predictor is Bayes-optimal. It predicts the posterior mean of $w^\top x$, so its error is the expected posterior variance of $w^\top x$. We write $R^{\mathrm{desc}}_{p=1}(n)$ for the value of $R^{\mathrm{desc}}(n)$ when the description is reliable. The ESS counts the further demonstrations that a predictor without the description would need to match the predictor with it.

Definition 1. Given n demonstrations, the effective sample size (ESS) of a description is $\mathrm{ESS}_n = n^* - n$, where n^* is the smallest $t \geq 0$ with $R^{\mathrm{demo}}(t) = R^{\mathrm{desc}}(n)$, with R^{demo} interpolated between the numbers of demonstrations at which it is known. If $R^{\mathrm{demo}}(t) < R^{\mathrm{desc}}(n)$ for every t, we write $\mathrm{ESS}_n \leq -n$: the description does more harm than all n demonstrations do good.

If $R^{\mathrm{desc}}(n)$ is below R^{demo} at every t measured, up to $t_{\max}$, we write $\mathrm{ESS}_n > t_{\max} - n$, and if above it at the smallest, $t_{\min}$, $\mathrm{ESS}_n \leq t_{\min} - n$; §6 estimates ESS_n from a sparse grid of t. The errors average over tasks, descriptions, and demonstrations, so the ESS belongs to a kind of description, set by r and p, and not to one value of m. For the Bayes-optimal predictor the ESS is never negative. Conditioning on the description cannot raise the expected posterior variance, so $R^{\mathrm{desc}}(n) \leq R^{\mathrm{demo}}(n)$, and R^{demo} decreases in n (Appendix A.1). For any other learner, the ESS is read from that learner's own two error curves (§§5–6). It then measures what the description is worth to that learner, and it is negative when the description hurts.

3. Specificity: Reliable Descriptions

We first consider reliable descriptions ($p = 1$). A reliable description narrows the prior over the task. Write $W_n = X_n^\top X_n$, with the n demonstration inputs as the rows of X_n. Without the description, the posterior covariance of w is $(I/a_0 + W_n)^{-1}$, and a reliable description replaces the baseline prior's precision $1/a_0$ with its own, c. Averaged over the query, the error is the trace of the posterior covariance:

$$R^{\mathrm{demo}}(n) = \mathbb{E}\,\mathrm{tr}(I/a_0 + W_n)^{-1},$$
$$R^{\mathrm{desc}}_{p=1}(n) = \mathbb{E}\,\mathrm{tr}(cI + W_n)^{-1}.$$

Proposition 1 (ESS of a reliable description)**.** *Let $p = 1$.*

(i) (Loose descriptions, worth about $(1-r)d$) $\mathrm{ESS}_0 \geq (1-r)\,d$*, and* $\mathrm{ESS}_0 \to (1-r)\,d$ *as $a_0 \to \infty$ with r fixed.*

(ii) (Specific descriptions, worth $c + d + 1$) If c is a whole number, $\mathrm{ESS}_0 \to c + d + 1$ *as $a_0 \to \infty$ with c fixed.*

We prove Proposition 1 in Appendix A.1 from two standard facts. With $n < d$ demonstrations, the data say nothing about $d - n$ directions, so $R^{\mathrm{demo}}(n) \geq (d-n)\,a_0$. With a flat baseline prior and $n \geq d+2$, $R^{\mathrm{demo}}(n) = d/(n-d-1)$, the trace of the mean of an inverse Wishart matrix (Muirhead, 1982). A reliable description leaves $R^{\mathrm{desc}}_{p=1}(0) = d/c = ra_0 d$, and the ESS is where these curves fall to that value.

Intuitively, the ESS has two parts. The first, the *dimension term*, counts dimensions. Each of the first d demonstrations removes the uncertainty in one direction. A loose description removes a fraction $1-r$ of the prior variance in every direction, so it is worth about $(1-r)d$ of them, and a specific one is worth about d. A specific description is also worth c, the ESS that Morita et al. (2008) give a normal prior, which is the ratio of the observation variance to the prior variance. Their ESS matches information, so a description that raises the prior precision from $1/a_0$ to c is worth $(1-r)\,c$ at every n. Ours matches errors instead, and n demonstrations leave the error $d/(n-d-1)$, not the d/n of their expected information. The dimension term is this gap, $d+1$ for a specific description, and it fades as demonstrations accumulate (Corollary 4(ii)).

Figure 2 shows the ESS as a function of r. At $d = 16$ and $a_0 = 10$, a description that leaves a tenth of the prior variance ($r = 0.1$) is worth 15.6 demonstrations, against $(1-r)d = 14.4$. One that leaves a thousandth ($r = 0.001$) is worth 117, which is $c + d + 1$. Demonstrations erode a description's value: with 100 in hand, the $r = 0.1$ description

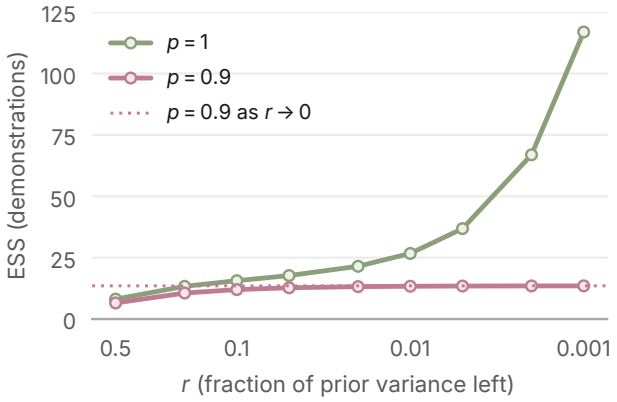


Figure 2. **Specificity raises the ESS of a reliable description without bound; one that may be irrelevant stays below a bound.** Bayes-optimal ESS of a description with no demonstrations in hand, against r, the fraction of the prior variance it leaves, for a reliable description ($p = 1$) and one that is relevant with probability 0.9. Dotted: the limit of the $p = 0.9$ ESS as $r \to 0$. $d = 16$, $a_0 = 10$; eight batches of 4,000 prompts, standard errors below 0.02 demonstrations.

is worth only 1.1.

Descriptions of part of the task. A description can be specific about some aspects of a task and silent about others. Let a reliable description leave the fraction r of the prior variance in $k < d$ directions and all of it in the other $d - k$. Its prior precision is then c in those k directions and $1/a_0$ in the rest.

Proposition 2 (ESS of a partial description)**.** *Let $p = 1$ and $k < d$.*

(i) (Without demonstrations, worth about k) As $a_0 \to \infty$, $\mathrm{ESS}_0 \to (1 - r)\, k$ with r fixed and $\mathrm{ESS}_0 \to k$ with c fixed.

(ii) (With many demonstrations, worth $(1 - r)ck/d$) As $n \to \infty$, $\mathrm{ESS}_n \to (1 - r)\, c\, k/d$.

Without demonstrations, a partial description is worth the directions it pins down, and its precision adds nothing. The error left in the directions it is silent about dwarfs what precision removes (Appendix A.3), so the $c + d + 1$ of Proposition 1(ii) needs a description of every direction. Once demonstrations have resolved the other directions, the description is worth the precision it adds, $k(1 - r)c$, in units of the information d that one demonstration carries. When this exceeds k, about when $(1 - r)c > d$, a partial description gains value as demonstrations accumulate. At $r = 0.001$, a description of half the directions is worth 8.1 demonstrations with none in hand and 36 with 100 (Table 1). Demonstrations thus substitute for a complete description but complement a partial one.

4. Reliability: Descriptions That May Be Irrelevant

We now turn to descriptions that may be irrelevant ($p < 1$) and show that, without demonstrations, their ESS is bounded however specific they are. Proposition 3 gives the price of not knowing whether a description is relevant, and the error in closed form when no demonstrations are in hand. When $p < 1$, the Bayes-optimal prediction mixes two predictions, $\mu_1^\top x$, which takes the description as relevant, and $\mu_0^\top x$, which ignores it. Here μ_1 and μ_0 are the posterior means of w under the two readings, and the weight is the posterior probability π_n that the description is relevant.

Proposition 3 (Error of a description that may be irrelevant)**.** *For every $n \geq 0$,*

$$R^{\mathrm{desc}}(n) \;=\; p\, R^{\mathrm{desc}}_{p=1}(n) + (1 - p)\, R^{\mathrm{demo}}(n) + \mathcal{V}_n, \quad (2)$$

where $\mathcal{V}_n = \mathbb{E}\big[\pi_n(1 - \pi_n)\big((\mu_1 - \mu_0)^\top x\big)^2\big] \geq 0$. With no demonstrations in hand,

$$R^{\mathrm{desc}}(0) = \big(1 - p^2(1 - r)\big)\, a_0 d. \quad (3)$$

The decomposition (2) is the law of total variance over Z (Appendix A.2). Its first two terms are the error of a *cheating* predictor, one that is told whether the description is relevant. With probability p this predictor holds a reliable description, and with probability $1 - p$ it holds only the demonstrations. To it, a description that may be irrelevant is worth what a reliable one is worth when it is relevant, and nothing when it is not.

The last term, $\mathcal{V}_n$, is the price of not being told: the variance of the prediction between the two readings of the description. It is small when the readings agree, or when the demonstrations make one reading far likelier than the other. Without demonstrations, it grows as a more specific description pulls the readings apart. With demonstrations, a specific description is also easier to tell from an irrelevant one. Equation (3) puts the two predictors side by side when no demonstrations are shown. Of the error $a_0 d$ that the baseline prior leaves, the cheating predictor removes a fraction $p(1 - r)$ and the Bayes-optimal predictor $p^2(1 - r)$, so not being told costs $p(1 - p)(1 - r)\, a_0 d$.

Corollary 4 (ESS of a description that may be irrelevant)**.**

(i) (Without demonstrations) If $p < 1$, then $\mathrm{ESS}_0 \leq n_p^{\max}$, where $n_p^{\max}$, which depends on d, a_0, and p but not on r, solves $R^{\mathrm{demo}}(n_p^{\max}) = (1 - p)\, R^{\mathrm{demo}}(0)$ for the interpolated R^{demo}. Also $\mathrm{ESS}_0 \geq p^2(1 - r)\, d$, and as $a_0 \to \infty$ with r fixed, $n_p^{\max} \to p\, d$ and $\mathrm{ESS}_0 \to p^2(1 - r)\, d$.

(ii) (With many demonstrations) As $n \to \infty$, $\mathrm{ESS}_n \to (1 - r)\, c$ if $p = 1$, and if $p < 1$,

$$\limsup_{n \to \infty} \mathrm{ESS}_n \leq p\,(1 - r)\, c.$$

Without demonstrations, the ESS is bounded. Refining a description from $r = 0.1$ to $r = 0.001$ raises its ESS from 15.6 to 117 if it is reliable, but only from 12.0 to 13.5 if it is relevant with probability 0.9 (Figure 2). However specific the description, the average error cannot fall below the share contributed by irrelevant descriptions, $(1 - p)R^{\mathrm{demo}}(0)$. Corollary 4(i) turns this floor into a bound on the ESS. At $p = 0.9$ the bound is $n_{0.9}^{\max} = 15.6$, the ESS of a reliable description with $r = 1-p$, which leaves the same error. The bound holds even for a predictor told whether the description is relevant, so it comes from irrelevant descriptions being worth nothing, not from uncertainty about relevance. That uncertainty lowers the ESS further, from about pd to about $p^2 d$ at a high signal-to-noise ratio.

With many demonstrations, the ESS is at most p times a reliable description's. Demonstrations shrink the error in the irrelevant case, so the ESS can grow. In the limit, a reliable description is worth $(1 - r)\, c$, the precision it adds to the baseline prior, and one that may be irrelevant is worth at most p times as much. With 100 demonstrations in hand, for example, the $r = 0.001$ description is worth 109 if it is reliable and 86.6 if it is relevant with probability 0.9.

Full trust can be worse than no description. A *fully trusting* predictor treats every description as relevant, predicting as if $p = 1$. With no demonstrations its error is

$$R^{\mathrm{trust}}(0) = \big(1 - (2p - 1)(1 - r)\big)\, a_0 d \qquad (4)$$

(Appendix A.4). It removes a fraction $(2p - 1)(1 - r)$ of the prior error, against $p^2(1 - r)$ for the Bayes-optimal predictor, and does worse than ignoring the description only when $p < 1/2$. Demonstrations do not rescue it. They show whether a specific description is relevant, but the fully trusting predictor keeps predicting near m, and as $r \to 0$ its error tends to $2(1 - p)\, a_0 d$ at every n. At $p = 0.9$ and $r = 0.001$, it is worse than no description from 15 demonstrations on.

5. Meta-Trained Transformers

To test whether a learner trained on the setting of §2 reaches the Bayes-optimal ESS, we meta-train Transformers that must learn from their training data how specific and how reliable descriptions are.

Unless specified otherwise, we use $d = 5$ and $a_0 = 10$, to keep training short. Prompts use the prefix embedding of Huang & Ge (2025): an optional description token carrying m, up to 20 demonstration tokens, and a query token. The network has the size of Garg et al. (2022)'s (12 layers, 8 heads, width 256) and uses their optimizer, but it is a bidirectional Transformer without positional encoding. Following Zhu et al. (2026), the network outputs a mean and a log variance for the query's label and is trained with log loss. Since the mean that minimizes this loss is the posterior mean, the Bayes-optimal prediction under squared error, we score the network's mean by its squared error (1). The description token carries only m, and half the training prompts omit it, so each network must learn r and p from its training distribution. We train a network for each specificity ($r = 0.1$, loose; $r = 0.01$, specific) and each reliability ($p \in \{1, 0.99, 0.9\}$), with three random seeds for the specific description and two for the loose one. For further details see Appendix B.

Comparing the networks to Bayes-optimal predictors. We test each network on fresh prompts at every n from 0 to 20, with and without a description, at its training reliability and at other reliabilities (Table 2). We read its ESS from its own error curves (Definition 1). We compare each network with two Bayes-optimal predictors: the *trained* one, which uses the reliability the network was trained on, and the *calibrated* one, which uses the reliability of the test prompts. To summarize a network's trust in a description, we define its *implied trust* as the q whose Bayes-optimal prediction with no demonstrations, $q\, m^\top x$, is closest to the network's prediction in squared error.

Networks come close to the Bayes-optimal ESS. Figure 3a shows networks trained and tested at the same reliability. For the specific description, the networks reach an ESS of 14.3 at $p = 1$, against a Bayes-optimal 16.1 on the same prompts. At $p = 0.99$ and $p = 0.9$ they come within 0.3 of the Bayes-optimal ESS (9.3 against 9.6, and 4.66 against 4.63), and for the loose description every network is within 0.1 of it (Table 2). The Bayes-optimal value at $p = 1$, 15.9 on a larger set of prompts, is close to the $c+d+1 = 16$ of Proposition 1(ii). This suggests that the networks learn to use a description nearly as well as the Bayes-optimal predictor does.

Training sets a network's trust. With no demonstrations, the prompt says nothing about relevance (Appendix A.2, Step 1), so any learner must take its trust from training. Figure 3b shows networks tested at every reliability. Networks trained at $p = 0.9$ keep hedging, with an implied trust near 0.9, even when every test description is relevant. Like the Bayes-optimal predictor trained at $p = 0.9$, they then fall well short of the calibrated ESS (Table 2). Networks trained on specific descriptions at $p = 1$ or 0.99 treat every description as relevant when no demonstrations are shown, but demonstrations separate them. When one test description in ten is irrelevant, the description still lowers the error of the $p = 0.99$ networks at every n, as it does the Bayes-optimal predictor's. The $p = 1$ networks trust it fully. As §4 predicts, this costs little without demonstrations, but from 6 demonstrations on their error is higher with the description than without, in every seed.

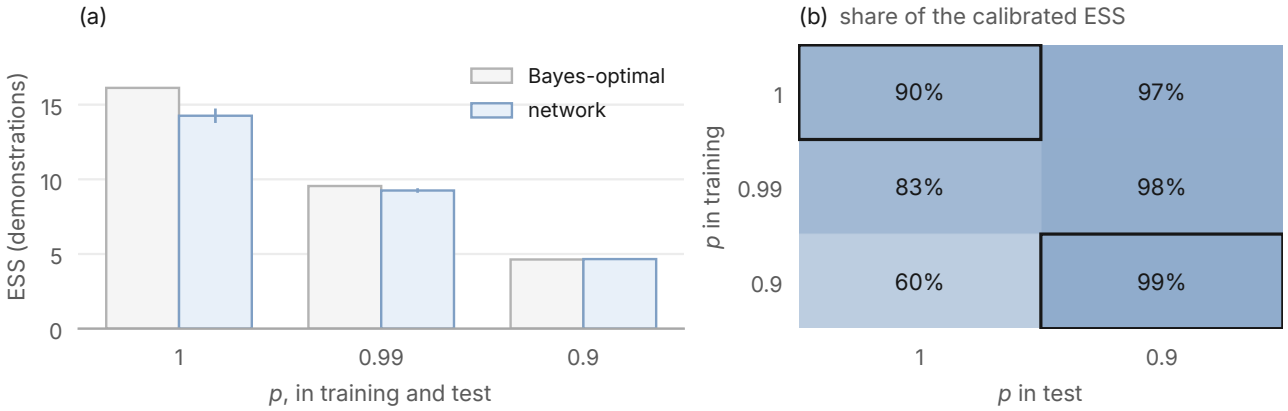


Figure 3. **Networks come close to the Bayes-optimal ESS and take their trust from training.** Specific description ($r = 0.01$), $d = 5$, $a_0 = 10$, no demonstrations; the network's ESS is its mean over three training seeds, with error bars of $\pm$ one standard deviation across them. (a) Trained and tested at the same reliability p: the Bayes-optimal ESS on the same prompts and the network's, read from the network's own no-description curve. (b) Trained at one reliability and tested at another: the network's ESS as a share of the calibrated predictor's; outlined, training matches test. Table 2 (Appendix B) gives every value, including those for the loose description ($r = 0.1$).

6. Pretrained Language Models

The networks of §5 learn how specific and how reliable descriptions are from their training data. A pretrained language model has no such training data for our descriptions. This raises a question: when the prompt states a description's specificity and reliability, does the model use the description as the Bayes-optimal predictor would? To probe this question, we state both properties in the prompt and compare the description's ESS for the model with its ESS for a predictor that is Bayes-optimal given what the prompt states. Following Gupta et al. (2025) and Akata et al. (2026), we compare against a known optimum, which a stated prior makes exact.

Unless specified otherwise, we study the simplest case of §2: one weight ($d = 1$) and a fixed input $x = 1$. A task is a hidden mean $w \sim \mathcal{N}(500, 100^2)$, and a demonstration is a number $y \sim \mathcal{N}(w, \sigma^2)$ with $\sigma = 60$, rounded to an integer. A description states a value m and a standard deviation $\tau \in \{60, 30, 10\}$, with $w \mid m \sim \mathcal{N}(m, \tau^2)$ ($r = \tau^2/100^2$), in one sentence of the prompt such as "Its mean was drawn from a normal distribution with mean 510 and standard deviation 30." We show up to 64 demonstrations, use 2,000 tasks, and repeat the test at $\sigma = 30$ as a check. We test six open chat models, answering without reasoning. gpt-oss-20b (OpenAI, 2025), Phi-4 (14B; Abdin et al., 2024), and OLMo 2 32B Instruct (Team OLMo et al., 2025) encode every three-digit integer as one token, so we take their prediction to be the mean of their first-token distribution over the 900 number tokens. Qwen3.5 9B (Qwen Team, 2026), Gemma 4 12B, and Gemma 4 31B (Gemma Team, 2026) split numbers into digits, so we take the number they write under greedy decoding. For further details see Appendix C.

Besides the relevant descriptions, we test three variations. An *irrelevant* description states the value that a relevant description of an independently drawn task would state. A *stated reliability* adds a sentence giving a 10% or 50% chance that the description is wrong, with the stated value and the demonstrations unchanged. A *specificity control* keeps the tasks, the demonstrations, and the stated value fixed and changes only the stated τ.

Comparing the models to the Bayes-optimal predictor. The predictor that is Bayes-optimal given exactly what the prompt says knows the noise σ and the description, with the mean otherwise uniform over the three-digit numbers. Given a relevant description and n demonstrations with mean $\bar{y}$, it predicts $(c\,m + n\,\bar{y})/(c+n)$ with $c = \sigma^2/\tau^2$. Intuitively, the description counts as c further demonstrations at m: the error is $\sigma^2/(c+n)$ with it and σ^2/n without, so its Bayes-optimal ESS is c at every n, up to the truncation of the uniform prior. Our values of τ give $c = 1$, 4, and 36, and we name the descriptions by c. We compare a model with this predictor in two ways. First, we read the ESS of Definition 1, with squared error against the hidden mean, off the model's own no-description error curve, interpolated in $\log n$ at its first crossing, and report the median over $n = 1$, 2, and 4. Read this way, the Bayes-optimal ESS is 1.3, 4.4, and 36. Where a model's own curve rises, we also read a *reference ESS* off the Bayes-optimal curve. Second, we regress the model's prediction on m, the demonstrations' mean, and a constant. The coefficient on m is the model's *weight* on the stated value, which is $c/(c+n)$ for the Bayes-optimal predictor.

Without reasoning, a specific description is worth a small fraction of its Bayes-optimal ESS. The Bayes-optimal ESS of the three relevant descriptions grows from 1 to 36

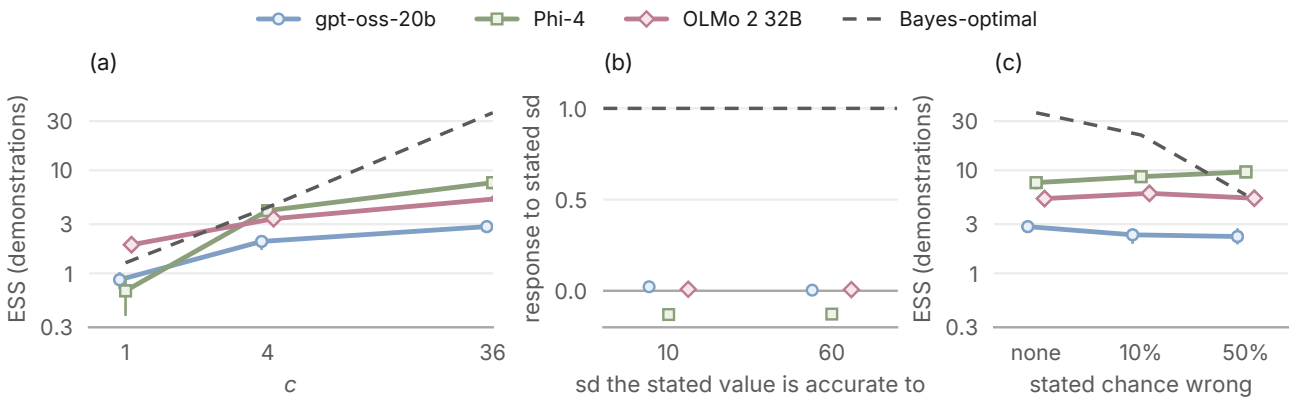


Figure 4. **Language models barely adjust to a description's stated specificity or reliability.** gpt-oss-20b, Phi-4, and OLMo 2 32B, read from their first-token distribution and answering without reasoning (the other three are in Table 3, Appendix C), against the Bayes-optimal predictor (dashed), for a relevant description; $\sigma = 60$, 2,000 tasks; error bars, 95% bootstrap intervals over tasks; in (a) and (c) the ESS is the median over $n = 1$, 2, and 4 demonstrations shown. (a) ESS of the descriptions with Bayes-optimal ESS $c = 1$, 4, and 36. (b) The specificity control (Table 7): the change in prediction when only the stated standard deviation goes from 60 to 10, as a share of the Bayes-optimal change over $n = 1$, 2, and 4, for a stated value accurate to standard deviation 10 or 60. (c) ESS of the $c = 36$ description with no stated chance of being wrong, or a stated 10% or 50% chance; the description is in fact relevant, and the Bayes-optimal predictor takes the stated chance at face value.

with c (Figure 4a), but the models' ESS grows far more slowly. The $c = 36$ description is worth a median of only 1.4 to 7.6 demonstrations across the six models (Table 3). One might expect that the models' ESS, though small, at least responds to the stated specificity. The specificity control shows that it largely does not. With the stated value held fixed, gpt-oss-20b's ESS stays near 2.7 whatever standard deviation the prompt states, while the Bayes-optimal ESS rises from 2.5 to 36 (Table 7). The models' ESS rises with c mainly because a more specific description states a value closer to the truth.

Without reasoning, the weight barely follows the stated specificity. The control also measures this response directly (Figure 4b). Changing only the stated standard deviation from 60 to 10 moves the Bayes-optimal weight by 0.47 at one demonstration. Across $n = 1$, 2, and 4, gpt-oss-20b and OLMo 2 32B move their prediction by at most 0.02 of the Bayes-optimal change, and Phi-4 moves it the other way. The three models read only from their written answer respond more. Gemma 4 31B responds the most, at about a third of the Bayes-optimal change and mostly at one demonstration. This indicates that the models under-use a specific description and over-use a loose one. Even at 64 demonstrations, gpt-oss-20b puts a weight of 0.23 on the $c = 1$ value, where the Bayes-optimal weight is 0.02, and the description raises its error.

Models discount a contradicted description. An irrelevant $c = 36$ description has a negative ESS for every model, but the models discount it. At four demonstrations, their errors (35 to 64) stay far below the 124 of the Bayes-optimal predictor, which takes the description as relevant.

A stated chance of being wrong does not move the ESS in proportion. Stating a 10% or 50% chance that the description is wrong lowers the Bayes-optimal ESS of the $c = 36$ description to 22 and 5.8 (Figure 4c). No model's ESS falls in this proportion (Table 3). gpt-oss-20b and OLMo 2 32B leave their ESS about where it was, and Phi-4's even rises. Qwen3.5 9B and Gemma 4 31B lower it about equally for both chances, and Gemma 4 12B's is higher at 50% than at 10%.

With reasoning, models follow the stated specificity. To test whether the models above lack the information or fail to use it when they answer at once, we rerun these conditions on 200 tasks with models that reason before answering: gpt-oss-20b at low effort, with enough new tokens that nearly all replies finish, and Claude Haiku 5.5 (Appendix C). Both now move their weight with the stated standard deviation about as much as the Bayes-optimal predictor does (Table 7). Haiku 5.5 also comes close to the Bayes-optimal ESS, 28 against 35 for the $c = 36$ description (Table 8), and GPT-5.6-Luna, reasoning inside the Codex agent, matches the Bayes-optimal weight and ESS (Appendix C). This suggests that the prompt conveys the stated specificity and that models use it when they reason first, although neither hedges for a stated 10% chance of being wrong.

A fitted weight can hide large errors. With reasoning, gpt-oss-20b gets the weight right, but about 2% of its answers are more than 40 from the Bayes-optimal prediction, from slips such as swapping the two variances. Its own no-

description curve rises after 16 demonstrations, so the ESS read from it exceeds the grid and says nothing. Read off the Bayes-optimal curve instead, the $c = 36$ description has a reference ESS of 24, against 35. The weight, an average response, barely registers these slips, but the reference ESS does not.

7. Related Work

ICL as Bayesian inference. Xie et al. (2022) cast ICL as implicit Bayesian inference. Transformers trained on samples from a prior approximate its posterior predictive (Müller et al., 2022), and meta-trained Transformers have been shown to implement Bayes-optimal estimators such as ridge regression under a Gaussian task prior, given enough task diversity (Akyürek et al., 2023; Raventós et al., 2023; Panwar et al., 2024).

Task descriptions in theories of ICL. Some theories of ICL include task descriptions, as tokens that indicate a hypothesis class (Lin et al., 2025) or carry the input mean (Huang & Ge, 2025), or as instructions in a generalization bound (Tong et al., 2026). Closest to our setting, Zhu et al. (2026) show that a Transformer given a prefix of related datasets performs amortized hierarchical Bayesian prediction. Our work values such conditioning information in demonstrations and allows it to be irrelevant.

Effective sample size of a prior. Our ESS follows Bayesian methods for the sample size of a prior (Morita et al., 2008; Neuenschwander et al., 2020; Reimherr et al., 2021). Like the ESS of Reimherr et al. (2021) and the effective current sample size of Wiesenfarth & Calderazzo (2020), it depends on the observations in hand, and, as in Reznik (2026), who discounts possibly mismatched historical data, it measures information by predictive loss. With many observations it returns to the ESS of Morita et al. (2008) (Corollary 4(ii)), but with few, matching errors rather than information adds a dimension term (§3). To model a description that may be irrelevant, we borrow the robust mixture prior that clinical trials use to discount conflicting historical data (Schmidli et al., 2014).

Descriptions and priors in pretrained language models. One instruction can stand in for up to 100 execution examples in fine-tuning data (Cook et al., 2026), and an optimized prompt can be shorter than demonstrations of equal effect (Wenliang et al., 2026). Tests against a known optimum show that in-context evidence outweighs a stated prior (Gupta et al., 2025) and that in-context labels override a model's own priors (Min et al., 2022; Wei et al., 2023). Learning curves approach a Bayesian reference (Akata et al., 2026; Arora et al., 2025), and text given alongside numbers shapes numerical predictions (Requeima et al., 2024; Liu et al., 2025). Our work adds a description with a stated specificity and reliability to such a test.

8. Discussion

Overall, for the Bayes-optimal predictor, specificity and task dimension raise a reliable description's ESS, a specific partial description gains value as demonstrations resolve the rest of the task, and a chance of irrelevance bounds the ESS without demonstrations. Meta-trained networks come close to these values and take their trust from training. Pretrained language models answering without reasoning fall far short of the Bayes-optimal ESS, and models that reason first follow the stated specificity, two of them coming close to it.

These results suggest that what a description is worth depends on the learner as well as on the description. Answering at once, most models give a stated value nearly the same weight whatever its stated specificity. Reasoning first, gpt-oss-20b weighs it nearly as the Bayes-optimal predictor does, although a few large errors still cost it much of its ESS. The ESS puts these behaviors in one unit.

Limitations. The analysis assumes a linear-Gaussian task and the robust mixture as the process generating descriptions. A relevant description errs only within its stated spread and an irrelevant one carries no information, whereas a real description can be partly right or more confident than it is accurate. The networks are trained at $d = 5$, and partial descriptions are untested. The language-model test covers only $d = 1$, with no dimension term, and states a numeric prior rather than an instruction. Three models are read from one greedy answer, and the reasoning runs use 200 tasks.

Impact Statement

This paper gives task descriptions a value in demonstrations and tests whether learners rely on them accordingly, which bears on how descriptions and demonstrations are combined in prompts. We see no societal consequences to discuss.

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A. Proofs

A.1. Proof of Proposition 1 (Reliable Descriptions)

Notation and standard facts. Throughout, $\varepsilon = 1/a_0$, $X_n \in \mathbb{R}^{n\times d}$ holds the demonstration inputs as rows, $W_n = X_n^\top X_n$, and for $\kappa > 0$

$$R^\kappa(n) = \mathbb{E}\,\mathrm{tr}(\kappa I + W_n)^{-1},$$

the error with n demonstrations under a Gaussian prior of precision κ in every direction, so that $R^{\mathrm{demo}} = R^\varepsilon$ and $R^{\mathrm{desc}}_{p=1} = R^c$. These are errors because the Bayes-optimal prediction is the posterior mean of $w^\top x$ and the query is independent of the rest of the prompt with $\mathbb{E}[xx^\top] = I$, so the error $\mathbb{E}[x^\top \Sigma x]$ is $\mathbb{E}\,\mathrm{tr}\,\Sigma$ for the posterior covariance Σ. We use three facts. (W1) For $n < d$, W_n has rank n almost surely, so $(\kappa I + W_n)^{-1}$ has the eigenvalue $1/\kappa$ with multiplicity $d - n$. (W2) For $n \geq d$ and a fixed unit vector e, $1/(e^\top W_n^{-1} e) \sim \chi^2_{n-d+1}$ (Muirhead, 1982, Theorem 3.2.12), so $\mathbb{E}\,\mathrm{tr}\,W_n^{-1} = d/(n-d-1)$ for $n \geq d+2$ and $\mathbb{E}\,\mathrm{tr}\,W_n^{-1} = \infty$ for $n \in \{d, d+1\}$. (W3) As $n \to \infty$ with d fixed, $\mathbb{E}\,\mathrm{tr}\,W_n^{-2} = d/n^2 + O(n^{-3})$ and $\mathbb{E}\,\mathrm{tr}\,W_n^{-3} = O(n^{-3})$ (von Rosen, 1988). By the Sherman–Morrison formula each new demonstration lowers $(\kappa I + W_n)^{-1}$ by a nonzero positive semidefinite matrix, so R^κ is strictly decreasing in n, and by (W2) it tends to 0.

Proof of Proposition 1. With the description and no demonstrations, $R^{\mathrm{desc}}_{p=1}(0) = d/c = ra_0 d$.

(i). By (W1), $R^\varepsilon(n) \geq (d-n)\,a_0$ for $n < d$, and trivially for $n \geq d$, so the interpolated R^{demo} satisfies $R^{\mathrm{demo}}(t) \geq (d-t)\,a_0$ at every real $t \geq 0$. For $t < (1-r)d$ this exceeds $ra_0 d$, so $\mathrm{ESS}_0 \geq (1-r)d$. As $a_0 \to \infty$, $R^\varepsilon(n)/a_0 = \mathbb{E}\,\mathrm{tr}(I + a_0 W_n)^{-1} \to (d-n)^+$ by dominated convergence, because each eigenvalue $1/(1+a_0\lambda)$ lies in $(0,1]$ and tends to 1 if $\lambda = 0$ and to 0 otherwise. So the interpolated $R^{\mathrm{demo}}(t)/a_0$ tends to $(d-t)^+$ at every t, a curve that falls through rd only at $t = (1-r)d$; as R^{demo} is decreasing, $\mathrm{ESS}_0 \to (1-r)d$.

(ii). With c fixed, $R^{\mathrm{desc}}_{p=1}(0) = d/c$ does not depend on a_0, while $R^\varepsilon(n)$ increases as a_0 grows and, by monotone convergence, (W1), and (W2), tends to $+\infty$ for $n \leq d+1$ (by (W1) for $n < d$ and (W2) for $n \in \{d, d+1\}$) and to $\mathbb{E}\,\mathrm{tr}\,W_n^{-1} = d/(n-d-1)$ for $n \geq d+2$. So the interpolated curve tends to $+\infty$ below $d+2$ and to the linear interpolation of $d/(n-d-1)$ above it, a strictly decreasing limit that, for whole c, equals d/c at $n = c + d + 1$; the root that defines ESS_0 therefore increases to $c + d + 1$. □

Extra precision with many demonstrations. Corollary 4(ii) needs the error to second order in $1/n$. Since $\kappa I + W_n$ and W_n commute,

$$(\kappa I + W_n)^{-1} = W_n^{-1} - \kappa\, W_n^{-2} + \kappa^2\, W_n^{-2}(\kappa I + W_n)^{-1},$$

and the last term lies between 0 and $\kappa^2 W_n^{-3}$. With (W2) and (W3), for fixed d and κ,

$$R^\kappa(n) = \frac{d}{n-d-1} - \frac{\kappa\, d}{n^2} + O(n^{-3}). \tag{5}$$

Each further demonstration lowers the first term by about d/n^2, and a prior of precision κ lowers the error by κ times as much, so once n is large it is worth κ demonstrations.

A.2. Proofs of Proposition 3 and Corollary 4 (Descriptions That May Be Irrelevant)

Let $D_n = (x_{1:n}, y_{1:n}, m, x)$ be the prompt with the query input, and let $\pi_n = P(Z = 1 \mid D_n)$ be the posterior probability that the description is relevant.

Step 1: the two readings. Given D_n and $Z = 1$, w is Gaussian with the posterior mean μ_1 and covariance Σ_1 of a reliable description. Given D_n and $Z = 0$, m is independent of w, and w is Gaussian with the posterior mean μ_0 and covariance Σ_0 from the demonstrations alone. At $n = 0$, $\pi_0 = p$: seeing m says nothing about Z, because m has the same law $\mathcal{N}(0, (1-r)a_0 I)$ in both cases, and x is independent of Z.

Step 2: the decomposition. The Bayes-optimal prediction is $\mathbb{E}[w^\top x \mid D_n] = (\pi_n \mu_1 + (1-\pi_n)\mu_0)^\top x$, and its error is $\mathbb{E}\,\mathrm{Var}(w^\top x \mid D_n)$. By the law of total variance over Z,

$$\mathrm{Var}(w^\top x \mid D_n) = \pi_n\, x^\top \Sigma_1 x + (1-\pi_n)\, x^\top \Sigma_0 x + \pi_n(1-\pi_n)\big((\mu_1 - \mu_0)^\top x\big)^2.$$

For any function g of the prompt, $\mathbb{E}[\pi_n\, g(D_n)] = \mathbb{E}[Z\, g(D_n)] = p\,\mathbb{E}[g(D_n) \mid Z = 1]$, and given $Z = 1$ the prompt has the law of a prompt with a reliable description, so $\mathbb{E}[\pi_n\, x^\top \Sigma_1 x] = p\, R^{\mathrm{desc}}_{p=1}(n)$. In the same way $\mathbb{E}[(1-\pi_n)\, x^\top \Sigma_0 x] = (1-p)\, R^{\mathrm{demo}}(n)$. This is (2).

Step 3: no demonstrations. At $n = 0$, $\pi_0 = p$, $\mu_1 = m$, and $\mu_0 = 0$, so the Bayes-optimal prediction is $p\, m^\top x$. In each coordinate, $\mathrm{Var}(w_i) = a_0$, $\mathrm{Var}(m_i) = (1-r)\, a_0$, and $\mathrm{Cov}(w_i, m_i) = p\,(1-r)\, a_0$, because w_i is m_i plus independent noise when the description is relevant and is independent of m_i when it is not. Hence

$$\mathbb{E}(w_i - p\, m_i)^2 = a_0 - 2p^2(1-r)\, a_0 + p^2(1-r)\, a_0 = \big(1 - p^2(1-r)\big)\, a_0.$$

The query is independent of (w, m) with $\mathbb{E}[xx^\top] = I$, so the error $\mathbb{E}\|w - p\, m\|^2$ is d times this, which is (3).

Proof of Corollary 4. (i). At $n = 0$, (2) with $\mathcal{V}_0 \geq 0$ gives $R^{\mathrm{desc}}(0) \geq (1-p)\, R^{\mathrm{demo}}(0)$ for every r, because $R^{\mathrm{desc}}_{p=1}(0) \geq 0$. Since R^{demo} is strictly decreasing and tends to 0 (Appendix A.1), there is a finite $n_p^{\max}$ with $R^{\mathrm{demo}}(n_p^{\max}) = (1-p)\, R^{\mathrm{demo}}(0)$, and the root that defines ESS_0 cannot exceed it. By (3), $R^{\mathrm{desc}}(0) = \big(d - p^2(1-r)\, d\big)\, a_0$, and $(1-p)\, R^{\mathrm{demo}}(0) = (d - pd)\, a_0$. The bound $R^{\mathrm{demo}}(t) \geq (d-t)\, a_0$ and the limit $R^{\mathrm{demo}}(t)/a_0 \to (d-t)^+$ from the proof of Proposition 1(i) then give $\mathrm{ESS}_0 \geq p^2(1-r)\, d$, $\mathrm{ESS}_0 \to p^2(1-r)\, d$, and $n_p^{\max} \to pd$.

(ii). Fix d, a_0, and r; every O below is as $n \to \infty$. Let $L(n) = p\, R^c(n) + (1-p)\, R^\varepsilon(n)$ be the first two terms of (2). By (5), uniformly over integers $0 \leq t \leq T = \lceil c \rceil + 1$,

$$R^{\mathrm{demo}}(n+t) - L(n) = \frac{d}{n^2}\Big[p\,(c-\varepsilon) - t + O(1/n)\Big].$$

The bracket is linear in t, so the same holds for the interpolated R^{demo} at real $t \in [0, T]$, and its root $n + \eta_n$, at which it equals $L(n)$, has $\eta_n = p\,(c-\varepsilon) + O(1/n)$, where $c - \varepsilon = (1-r)\, c$. If $p = 1$, then $R^{\mathrm{desc}} = L$ and $\mathrm{ESS}_n = \eta_n \to (1-r)\, c$. If $p < 1$, then $R^{\mathrm{desc}}(n) = L(n) + \mathcal{V}_n \geq L(n)$ and R^{demo} is decreasing, so $\mathrm{ESS}_n \leq \eta_n$ and $\limsup_n \mathrm{ESS}_n \leq p\,(1-r)\, c$. □

A.3. Proof of Proposition 2 (Partial Descriptions)

Let $\Lambda = \mathrm{diag}(cI_k, \varepsilon I_{d-k})$, so that the error with the description is $\mathbb{E}\,\mathrm{tr}(\Lambda + W_n)^{-1}$. The law of W_n is invariant under rotations, so it does not matter which k directions the description informs.

Proof of Proposition 2. (i). Without demonstrations the error with the description is $\mathrm{tr}\,\Lambda^{-1} = (kr + d - k)\, a_0$. As in the proof of Proposition 1(i), $R^{\mathrm{demo}}(t) \geq (d-t)\, a_0$ at every $t \geq 0$, so $\mathrm{ESS}_0 \geq (1-r)k$, and $R^{\mathrm{demo}}(t)/a_0 \to (d-t)^+$, uniformly in t because the curves are monotone and the limit is continuous. With r fixed, the error with the description over a_0 is $d - (1-r)k$; with c fixed, it is $d - k + k/(c\, a_0) \to d - k$. As $k < d$, the limit curve passes through each value at a single point, $(1-r)k$ or k, with slope -1, so the root that defines ESS_0 converges to that point.

(ii). Expanding twice,

$$(\Lambda + W_n)^{-1} = W_n^{-1} - W_n^{-1}\Lambda W_n^{-1} + W_n^{-1}\Lambda W_n^{-1}\Lambda(\Lambda + W_n)^{-1}.$$

By rotation invariance, $\mathbb{E}W_n^{-2} = (\mathbb{E}\,\mathrm{tr}\, W_n^{-2}/d)\, I$, so by (W3) the middle term has expected trace $\mathrm{tr}(\Lambda)/n^2 + O(n^{-3})$. For the last term, $|\,\mathrm{tr}(MB)| \leq \|M\|\,\mathrm{tr}\, B$ for $B \succeq 0$; with $B = (\Lambda + W_n)^{-1} \preceq W_n^{-1}$ its trace is at most $\|\Lambda\|^2 \|W_n^{-1}\|^2\,\mathrm{tr}\, W_n^{-1} \leq d\,\|\Lambda\|^2\,\mathrm{tr}\, W_n^{-3}$, whose expectation is $O(n^{-3})$ by (W3). Hence

$$\mathbb{E}\,\mathrm{tr}(\Lambda + W_n)^{-1} = \frac{d}{n - d - 1} - \frac{\mathrm{tr}\,\Lambda}{n^2} + O(n^{-3}),$$

which is (5) with κd replaced by $\mathrm{tr}\,\Lambda$. The proof of Corollary 4(ii) with $p = 1$ then gives $\mathrm{ESS}_n \to (\mathrm{tr}\,\Lambda - d\varepsilon)/d = k(c-\varepsilon)/d = (1-r)\, c\, k/d$. □

The same steps give the limit $(\mathrm{tr}\,\Lambda - d\varepsilon)/d$ for a reliable description with any Gaussian prior precision Λ: with many demonstrations in hand, a description is worth the precision it adds averaged over directions.

Table 1. **A specific partial description gains value as demonstrations accumulate.** Bayes-optimal ESS of a reliable description of k of the $d = 16$ directions ($k = 16$: a complete description), $a_0 = 10$, with n demonstrations in hand; the last column of each block is the limit of Proposition 2(ii), $(1 - r)ck/d$. Eight batches of 2,000 prompts; standard errors at most 0.23.

	loose, $r = 0.1$ ($c = 1$)					specific, $r = 0.001$ ($c = 100$)				
k	$n = 0$	$n = 10$	$n = 100$	$n = 400$	$n \to \infty$	$n = 0$	$n = 10$	$n = 100$	$n = 400$	$n \to \infty$
1	0.9	0.8	0.1	0.1	0.1	1.0	1.0	3.5	5.2	6.2
4	3.6	2.9	0.3	0.2	0.2	4.0	4.2	15.5	21.7	25.0
8	7.3	5.1	0.5	0.5	0.5	8.1	9.8	36.2	46.0	50.0
12	11.1	6.7	0.8	0.7	0.7	12.4	23.4	65.1	73.0	74.9
16	15.6	8.0	1.1	0.9	0.9	117	116	109	103	99.9

A.4. Derivation of Equation (4) (The Fully Trusting Predictor)

With no demonstrations the fully trusting predictor predicts $m^\top x$. With the moments of Step 3, $\mathbb{E}(w_i - m_i)^2 = a_0 - 2p(1 - r)\, a_0 + (1 - r)\, a_0 = \big(1 - (2p - 1)(1 - r)\big)\, a_0$, and summing over the d coordinates gives (4); it exceeds $R^{\mathrm{demo}}(0) = a_0 d$ exactly when $p < 1/2$. With n demonstrations it predicts $\mu_1^\top x$ whatever Z, where $\mu_1 = m + (cI + W_n)^{-1} X_n^\top (y_{1:n} - X_n m)$. As $r \to 0$ with a_0 and n fixed, $c \to \infty$ and $\mu_1 \to m$ in mean square, so its error tends to $\mathbb{E}\|w - m\|^2$: $r a_0 d \to 0$ for a relevant description and $(2 - r)\, a_0 d \to 2 a_0 d$ for an irrelevant one, $2(1 - p)\, a_0 d$ in all. Since $R^{\mathrm{demo}}(n) \to 0$, for every $p < 1$ there is an n from which full trust does worse than no description once r is small enough.

B. The Meta-Trained Networks in Full

Table 2. Networks against the reference predictors ($d = 5$, $a_0 = 10$; 20,000 evaluation prompts per network), grouped by training reliability. "Network" is the mean ESS over the training seeds, ± the standard deviation across them (for one seed, the standard error over prompts); "trust" is also averaged over seeds. "Trained" and "calibrated" are the Bayes-optimal predictors with the training and the test reliability, shown once where they coincide. Calibrated values, and trained values where the two reliabilities coincide, use 200,000 prompts (standard errors at most 0.1); other trained values use the network's evaluation prompts, on which prompt noise alone can move them by up to 0.6, so a trained value can exceed the calibrated one. "Trust" is the network's implied trust with no demonstrations.

				ESS			
r	p_{train}	p_{test}	Seeds	Network	Trained	Calibrated	Trust
0.1	1	1	2	5.85 ± 0.03	5.89	—	0.964
0.1	1	0.99	1	5.65 ± 0.06	5.66	5.61	0.991
0.1	1	0.9	2	4.02 ± 0.03	3.98	4.10	0.964
0.01	1	1	3	14.26 ± 0.48	15.91	—	1.008
0.01	1	0.99	2	9.38 ± 0.16	9.54	8.96	1.015
0.01	1	0.9	3	4.58 ± 0.03	4.54	4.70	1.008
0.1	0.99	1	2	5.87 ± 0.01	5.88	5.89	0.963
0.1	0.99	0.99	2	5.65 ± 0.01	5.61	—	0.963
0.1	0.99	0.9	2	4.02 ± 0.03	3.99	4.10	0.963
0.01	0.99	1	3	13.28 ± 0.57	16.01	15.91	0.982
0.01	0.99	0.99	3	9.25 ± 0.15	8.96	—	0.982
0.01	0.99	0.9	3	4.60 ± 0.04	4.56	4.70	0.982
0.1	0.9	1	2	5.67 ± 0.06	5.75	5.89	0.883
0.1	0.9	0.99	1	5.51 ± 0.06	5.55	5.61	0.901
0.1	0.9	0.9	2	4.03 ± 0.01	4.10	—	0.883
0.01	0.9	1	3	9.49 ± 0.44	10.79	15.91	0.876
0.01	0.9	0.99	2	8.08 ± 0.28	8.55	8.96	0.880
0.01	0.9	0.9	3	4.66 ± 0.01	4.70	—	0.876

Training details. Each demonstration token holds an input beside its label, and two indicator coordinates mark the token type; there is no positional encoding. The Transformer has 12 layers, 8 heads, and width 256. It is trained with Adam (learning rate 10^{-4}) for 40,000 steps at batch size 1,024, on fresh prompts at every step and without a curriculum. Without a description, the description token is masked out of attention, and the prediction is read at the query token. The labels and the description enter divided by $\sqrt{a_0}$; gradients are clipped to norm 1, and training uses bfloat16 autocast. The number of

demonstrations in a training prompt is uniform on 0 to 20 (0 to 15 for the $r = 0.05$ and $r = 0.5$ networks below). Four further networks, at $r = 0.5$ ($p = 1$ and 0.9) and $r = 0.05$ ($p = 1$ and 0.9), reach an ESS within 0.03 of the Bayes-optimal ESS (2.64 against 2.66, 2.16 against 2.15, 7.32 against 7.35, and 4.38 against 4.37). Without a description, every network's error is within 0.21 of the Bayes-optimal predictor's (about 50) at $n = 0$ and within 0.3 on average over the numbers of demonstrations it was trained on.

ESS as demonstrations accumulate. Figure 5 gives each network's ESS_n at its training reliability. For the loose description it follows the Bayes-optimal ESS_n to within 0.6 at every n the grid can show. For the specific one it falls short by up to 2.4, 2.9, and 1.4 demonstrations at $p = 1$, 0.99, and 0.9, and exceeds it by at most 0.3, until the crossing leaves the grid; with five demonstrations in hand, the three $p = 0.9$ networks have an ESS of 4.83, 4.79, and 4.77, against 5.20.

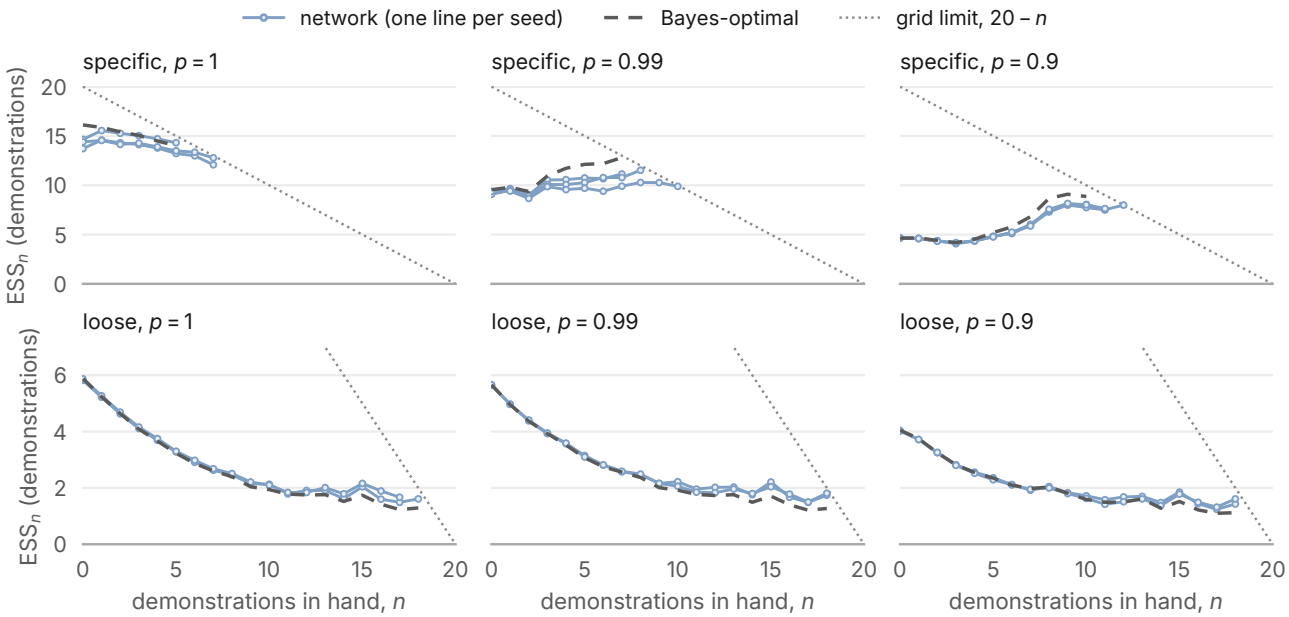


Figure 5. **The networks' ESS as demonstrations accumulate.** ESS_n of each meta-trained network (one line per training seed) and of the Bayes-optimal predictor on the same prompts, trained and tested at the same reliability p, for the specific ($r = 0.01$) and the loose ($r = 0.1$) description; $d = 5$, 20,000 prompts. A curve stops where the crossing lies beyond the 20 demonstrations of the grid; the dotted line, $20 - n$, is the largest ESS the grid can show.

C. The Language-Model Test in Full

Table 3 gives the headline values with their intervals for the six models (gpt-oss-20b under both readouts). Figure 6 shows the error curves the ESS is read from, Tables 11 and 12 the ESS of every description at every number of demonstrations shown, Table 4 the headline ESS under other readings of the curves, Table 13 the weight on the stated value at every number of demonstrations shown, for the three models read from their distribution, and Table 5 the ESS under actual unreliability.

Models. MiniCPM5 2B (MiniCPM Team, 2026), OLMo 2 7B Instruct, and OLMo 2 13B Instruct begin only 62%, 20%, and 10% of their replies with a number, often explaining instead ("To predict…", "Given…") despite the instruction to reply with the number only, and are not reported beyond that (OLMo 2 7B's $c = 36$ ESS reads 21, from a readout resting on a minority of its probability); we report a model only where it begins at least three quarters of its replies with a number overall, and at least half of its no-description replies over $n = 1$, 2, and 4, the curve its ESS is read from. Thinking is disabled where the chat template allows it, and we start gpt-oss-20b's reply in its final channel. The models run in Hugging Face Transformers at their native precision (`openai/gpt-oss-20b`, `microsoft/phi-4`, `allenai/OLMo-2-0325-32B-Instruct`, `Qwen/Qwen3.5-9B`, `google/gemma-4-12B-it`, `google/gemma-4-31B-it`), and tasks come from a seeded generator.

Prompt. The user turn of the model's chat template:

Table 3. The ESS and the weight on the stated value for each model. Left: the ESS of a description (median over $n = 1$, 2, and 4 demonstrations shown) for a relevant description with $c = 1$, 4, or 36, and for the $c = 36$ description when irrelevant or with a stated 10% or 50% chance of being wrong. Brackets, 95% bootstrap intervals; $\leq$, an upper bound, where one demonstration alone beats the description; $\sigma = 60$, 2,000 tasks; "written": read from the number the model writes. Right: the weight on the stated value with one demonstration shown; the Bayes-optimal weight is $c/(c+n)$. The Bayes-optimal predictor takes a description as relevant unless a chance of being wrong is stated, and then hedges with that chance.

	ESS, median over $n = 1, 2, 4$						weight, $n = 1$		
	relevant			$c = 36$			relevant		
Model	$c = 1$	$c = 4$	$c = 36$	irrelevant	stated 10%	stated 50%	$c = 1$	$c = 4$	$c = 36$
Bayes-optimal	1.3	4.4	36	≤ -1.0	22	5.8	0.50	0.80	0.97
gpt-oss-20b	0.9 [0.7, 1.0]	2.0 [1.7, 2.3]	2.8 [2.6, 3.1]	≤ -0.4	2.4 [1.9, 2.6]	2.3 [1.9, 2.7]	0.53	0.48	0.49
Phi-4	0.7 [0.4, 0.9]	4.1 [3.5, 4.6]	7.6 [6.5, 8.6]	≤ -1.0	8.7 [7.8, 9.8]	9.7 [8.7, 11]	0.82	0.76	0.74
OLMo 2 32B	1.9 [1.7, 2.1]	3.4 [3.2, 3.9]	5.3 [4.5, 6.3]	≤ -0.6	6.0 [5.6, 7.1]	5.4 [5.0, 6.0]	0.72	0.70	0.70
gpt-oss-20b (written)	0.2 [0.1, 0.4]	2.0 [1.9, 2.6]	3.7 [3.0, 4.4]	≤ -1.0	2.4 [2.1, 2.6]	2.3 [2.1, 2.6]	0.64	0.52	0.52
Qwen3.5 9B (written)	0.2 [−0.1, 0.4]	2.7 [2.1, 3.2]	4.0 [3.5, 4.5]	≤ -1.0	0.6 [0.3, 1.0]	1.0 [0.6, 1.5]	0.81	0.76	0.80
Gemma 4 12B (written)	0.0 [−0.1, 0.1]	0.5 [0.3, 0.7]	1.8 [1.4, 2.2]	≤ -0.7	1.3 [1.1, 1.6]	2.7 [2.5, 3.1]	0.34	0.24	0.59
Gemma 4 31B (written)	0.0 [0.0, 0.0]	0.2 [0.1, 0.3]	1.4 [1.2, 1.6]	≤ -0.7	0.6 [0.5, 0.7]	0.5 [0.4, 0.6]	0.48	0.49	0.72

```
The numbers below are drawn from a normal distribution with standard deviation 60.
Its mean was drawn from a normal distribution with mean 510 and standard deviation 30.
548
431
Predict the next number.  Reply with the number only.
```

Without a description, the second sentence is "Its mean is unknown." With a stated reliability, a third sentence follows: "There is a 10% chance that this statement is wrong, in which case the mean is unknown." With no demonstrations, the question is "Predict the first number." The stated value of a relevant description is drawn given the task, so that $m \sim \mathcal{N}(500, 100^2 - \tau^2)$ and $w \mid m \sim \mathcal{N}(m, \tau^2)$; a task is redrawn in the rare case (under 1%) that a number or stated value falls outside 100 to 999.

Measurement. The ESS of §6 is Definition 1 with squared error. Its summary leaves out $n = 0$, where without a description the models rarely answer with a number, and large n, where the ESS mixes the description's value with the models' slow learning (error 9 to 23 after 64 demonstrations at $\sigma = 60$, against 7.5 for the Bayes-optimal predictor) and their over-reliance on loose descriptions. A model that learns slowly from demonstrations values a description more in its own units; the weights do not depend on this. The no-description mean squared error at $n = 1, 2, 4, \ldots, 64$ is interpolated linearly in $\log n$, on the squared error, not its root; ESS_n is the first t at which that curve falls to the description's mean squared error at n, minus n. Since the curve starts at one demonstration, when one demonstration alone beats the description we report the bound $\mathrm{ESS}_n \leq 1 - n$. Each bootstrap replicate resamples tasks, the same tasks for both curves, and repeats the reading, so the intervals reflect sampling over tasks under this procedure, not the interpolation bias quantified next.

The Bayes-optimal reference. The Bayes-optimal predictor is that of §6, with the mean uniform over 100 to 999, so that without a description its prediction is essentially the mean of the demonstrations shown (the range barely truncates it). A relevant description's exact Bayes-optimal ESS is then c, up to that truncation, that is 1, 4, and 36. Computed the same way as the models' ESS, it reads 1.3, 4.4, and 36 at $\sigma = 60$, and 1.2, 4.7, and 43 at $\sigma = 30$: the $\log n$ interpolation biases it upward (by about 0.2, 0.3, and 1.8 for $c = 1$, 4, and 36 with exact errors), and the finite sample of tasks shifts it further, in either direction (for $c = 36$, down to 36 at $\sigma = 60$ and up to 43 at $\sigma = 30$). The tables report the computed value for the Bayes-optimal predictor and the models alike. Under the prior the tasks are drawn from, $\mathcal{N}(500, 100^2)$, a predictor without the description would hold a prior precision of $\sigma^2/100^2 = 0.36$ demonstrations at $\sigma = 60$ (0.09 at $\sigma = 30$), and a relevant description's exact ESS would be $c - 0.36$, that is 0.64, 3.64, and 35.64, the $(1 - r)\,c$ of §3. We keep the predictor that knows only what the prompt states, since the models are told nothing else.

Reading the curves. Definition 1 interpolates between the numbers of demonstrations at which the error is known, and the grid here doubles n, so how the curve is read between grid points matters. On the Bayes-optimal predictor's exact errors, σ^2/n without the description and $\sigma^2/(c+n)$ with it, interpolation in $\log n$ overstates the ESS by 0.2, 0.3, and 1.8

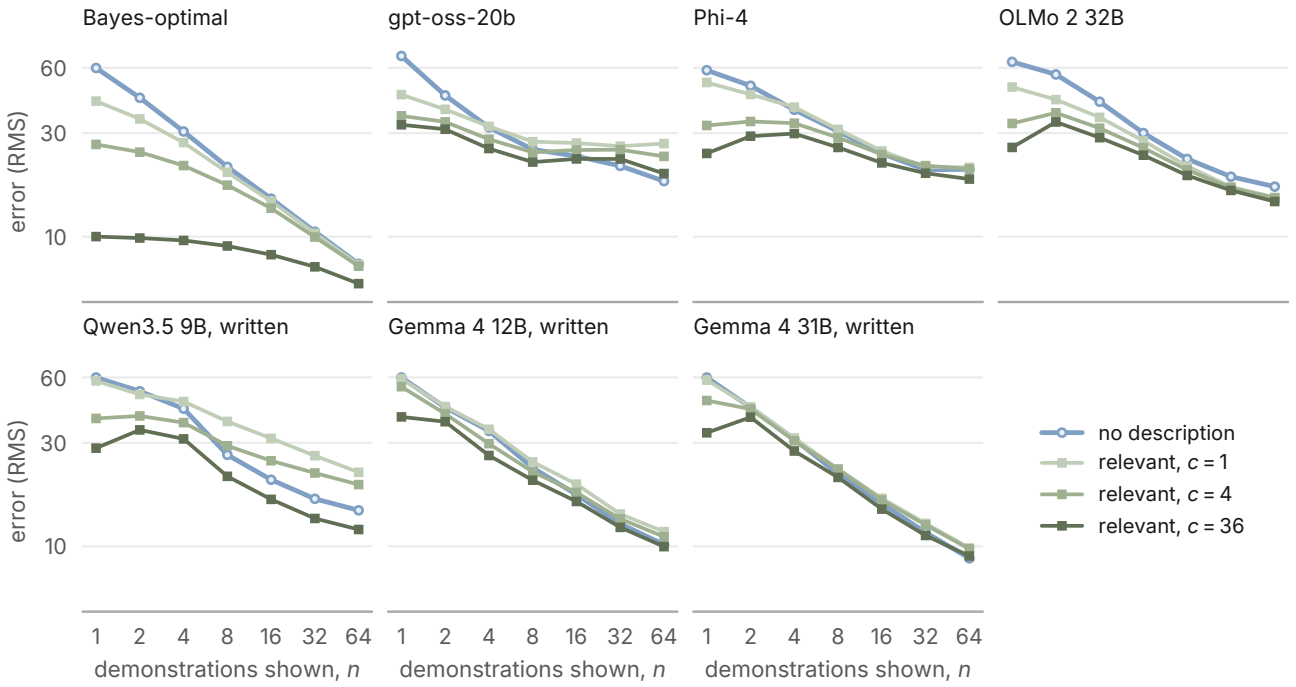


Figure 6. **The error curves the language-model ESS is read from.** Root-mean-square error against the number of demonstrations shown, without a description (blue) and with a relevant description (green, darker for larger c), for the Bayes-optimal predictor and the six reported models; $\sigma = 60$, 2,000 tasks, replies without a number left out. The ESS of a description at n is the horizontal distance from its point at n to the blue curve.

for $c = 1$, 4, and 36, and interpolation in n by 0.3, 0.6, and 4.1. For models answering without reasoning the first crossing is also the only one: of their twelve no-description curves at the two noise levels, only Phi-4's at $\sigma = 60$ rises, from 20.3 at 32 demonstrations to 20.4 at 64. Table 4 gives the headline ESS under four readings: ours; interpolation in n; the curve made non-increasing by a least-squares monotone fit (pool adjacent violators); and the curve $A/(n + \kappa) + B$ fitted to the no-description errors by relative least squares, the form of a Bayesian learner's error plus a floor, which recovers the exact ESS from the Bayes-optimal predictor's exact errors. No model's headline ESS moves by more than 1.4 demonstrations, and none exceeds 8.1 for the $c = 36$ description, against 35 to 38 for the Bayes-optimal predictor. The readings differ more at larger n, where a model's no-description curve flattens and a small difference in error spans many demonstrations: gpt-oss-20b's $c = 36$ ESS at $n = 8$ is 17 in Table 11 and 9.2 from the fitted curve. In Figure 6, the error of four models with the $c = 36$ description rises from one demonstration shown to two, because each demonstration pulls their prediction off the stated value much more than it pulls the Bayes-optimal prediction (Table 13).

Replies. A model's share of replies that begin with a number is the probability its first token puts on the number tokens; where that share is below three quarters, Table 13 sets the cell in italics. The noise-30 conditions, which realize the same three c with both standard deviations in the prompt halved, show the same pattern as the noise-60 ones (Table 11). OLMo 2 32B's first token puts 41% of its probability on numbers in the no-description prompts with one demonstration shown and 75% with two, so its no-description prediction at $n = 1$ rests on a minority of that probability.

Written readout. For Qwen3.5 9B, Gemma 4 12B, and Gemma 4 31B, and as a check on the others, the prediction is the first number with three digits before any decimal part in the model's reply under greedy decoding with eight new tokens, decimals kept, the rule for every text reply here (these runs kept only the integer part, but direct answers rarely have a decimal part: 3 of GPT-5.6-Luna's 1,300 with reasoning off). A reply without one is left out of both the error curves and the weight regression; the share of replies with one is the "number" column. On gpt-oss-20b the written readout gives weights within 0.21 of the distribution readout's in every condition and at every n, and headline ESS medians of 0.2, 2.0, and 3.7 against 0.9, 2.0, and 2.8; it is the noisier readout, since one greedy answer stands in for the distribution's mean. Phi-4 begins only 63% of its written replies with a number over all conditions, so Table 3 reads it from its distribution only; in the wording ablation ($n = 1$, 2, 4) its share is above three quarters for the paper's wording. OLMo 2 32B writes a number

Table 4. **The headline ESS under four readings of the no-description curve.** Median over $n = 1$, 2, and 4 demonstrations shown for the three relevant descriptions; $\sigma = 60$, 2,000 tasks. Log n: the readings of Table 3 (first crossing, interpolation in $\log n$); linear: interpolation in n; monotone: after a monotone fit; fit: the fitted curve $A/(n + \kappa) + B$. The exact Bayes-optimal ESS is 1, 4, and 36.

	$c = 1$				$c = 4$				$c = 36$			
Predictor	log n	linear	monotone	fit	log n	linear	monotone	fit	log n	linear	monotone	fit
Bayes-optimal	1.3	1.4	1.3	1.0	4.4	4.7	4.4	4.0	36	38	36	35
gpt-oss-20b	0.9	1.0	0.9	0.9	2.0	2.4	2.0	2.4	2.8	2.9	2.8	3.4
Phi-4	0.7	0.8	0.7	0.4	4.1	4.4	4.1	3.6	7.6	8.1	7.6	7.1
OLMo 2 32B	1.9	2.1	1.9	1.4	3.4	3.7	3.4	3.3	5.3	5.8	5.3	5.5
Qwen3.5 9B (written)	0.2	0.3	0.2	−0.1	2.7	2.9	2.7	1.3	4.0	4.3	4.0	2.8
Gemma 4 12B (written)	0.0	0.0	0.0	0.0	0.5	0.6	0.5	0.4	1.8	1.9	1.8	1.6
Gemma 4 31B (written)	0.0	0.0	0.0	−0.1	0.2	0.3	0.2	0.0	1.4	1.7	1.4	1.0

Table 5. **ESS under actual unreliability.** The relevant and irrelevant conditions combined with weights p and $1 - p$; median over $n = 1$, 2, and 4 demonstrations shown, with 95% bootstrap intervals over tasks; $\sigma = 60$, 2,000 tasks. A bound where one demonstration alone beats the description at some n.

	irrelevant one time in ten ($p = 0.9$)		irrelevant half the time ($p = 0.5$)	
Predictor	$c = 4$	$c = 36$	$c = 4$	$c = 36$
Bayes-optimal, takes it as relevant	1.0 [0.8, 1.1]	0.2 [0.0, 0.4]	≤ -1.0	≤ -1.0
Bayes-optimal, hedges at p	2.3 [2.1, 2.7]	5.5 [5.1, 6.0]	0.6 [0.5, 0.7]	0.9 [0.8, 1.1]
gpt-oss-20b, not told	1.2 [1.0, 1.4]	1.6 [1.4, 1.7]	0.1 [0.0, 0.2]	≤ 0.0
gpt-oss-20b, told	1.1 [1.0, 1.3]	1.3 [1.1, 1.4]	−0.1 [−0.1, 0.0]	−0.1 [−0.1, 0.0]
Phi-4, not told	1.3 [1.1, 1.5]	1.9 [1.7, 2.4]	≤ -1.0	≤ -1.0
Phi-4, told	2.7 [2.2, 3.2]	3.6 [3.1, 4.1]	≤ 0.0	≤ 0.0
OLMo 2 32B, not told	2.5 [2.2, 2.8]	3.2 [2.9, 3.5]	≤ 0.7	≤ 0.6
OLMo 2 32B, told	3.2 [2.9, 3.4]	4.0 [3.6, 4.3]	≤ 1.0	≤ 0.9
Qwen3.5 9B (written), not told	1.0 [0.7, 1.2]	1.8 [1.3, 2.1]	≤ -0.3	≤ -0.9
Qwen3.5 9B (written), told	−1.0 [−1.0, −0.7]	−0.1 [−0.5, 0.3]	≤ -1.0	≤ -1.0
Gemma 4 12B (written), not told	0.3 [0.2, 0.5]	0.7 [0.5, 1.0]	≤ -0.1	≤ 0.0
Gemma 4 12B (written), told	0.6 [0.5, 0.7]	1.0 [0.9, 1.3]	≤ 0.0	≤ 0.1
Gemma 4 31B (written), not told	0.1 [0.0, 0.2]	0.4 [0.3, 0.5]	≤ 0.0	≤ -0.1
Gemma 4 31B (written), told	≤ 0.0	0.2 [0.1, 0.3]	≤ 0.0	≤ 0.0

in only 5%, 36%, and 48% of its no-description replies at $n = 1$, 2, and 4 ($\sigma = 60$), so its written ESS is not reported; its written weights on the three relevant descriptions at $n = 1$ are 0.83, 0.81, and 0.83, against 0.72, 0.70, and 0.70 from its distribution.

Actual unreliability. The stated-reliability conditions keep the description relevant and test whether a model hedges as told. Table 5 instead realizes the mixture of §2: the squared errors of the relevant and the irrelevant condition, on the same tasks, are combined with weights p and $1 - p$ before the ESS is read, for a model not told the chance and for one told it, against the fully trusting Bayes-optimal predictor and the calibrated one, which hedges at p. With one description in ten irrelevant, the $c = 36$ description is worth 5.5 demonstrations to the calibrated predictor and 0.2 to the fully trusting one; to the models it is worth 0.4 to 3.2 when not told and −0.1 to 4.0 when told. Their flat weight on a description acts as a hedge between full trust and calibration, and being told the chance moves it by at most two demonstrations, in either direction. With half the descriptions irrelevant, the calibrated predictor keeps under one demonstration and no model keeps more than one.

Wording of the description. We reran the three relevant descriptions and the no-description prompt at $n = 1$, 2, and 4 with three other wordings of the description sentence: "Its mean is about m, give or take τ.", "Mean: $m \pm \tau$", and "An earlier estimate put its mean at m, with a standard error of τ." (Table 6; written readout, 2,000 tasks). The wording sets how much a model relies on the description: with one demonstration shown, Gemma 4 12B's weight on the $c = 1$ value is 0.34 under the paper's wording and 1.00 under the other three, and gpt-oss-20b's ranges from 0.17 to 0.67 (a weight above 1 means the prediction moves more than the stated value does). Under no wording does the weight follow the stated specificity

as the Bayes-optimal weight does: the gap between $c = 36$ and $c = 1$ is at most 0.25 at $n = 1$ and 0.44 at $n = 4$ over all models and wordings, against 0.47 and 0.70 for the Bayes-optimal predictor.

Table 6. **The wording of the description sets how much a model relies on it, but under no wording does the weight follow the stated specificity.** Weight on the stated value for the three relevant descriptions at $n = 1$ and 4 under four wordings (written readout, $2{,}000$ tasks). Number: the share of replies with a number; italics: below three quarters in that cell. The Bayes-optimal weight is 0.50, 0.80, and 0.97 at $n = 1$ and 0.20, 0.50, and 0.90 at $n = 4$.

			weight, $n = 1$			weight, $n = 4$		
Model	Wording of the description	number	$c = 1$	$c = 4$	$c = 36$	$c = 1$	$c = 4$	$c = 36$
Gemma 4 12B	normal, mean m, sd τ (the paper's)	1.00	0.34	0.24	0.59	0.19	0.24	0.31
	about m, give or take τ	1.00	1.00	1.00	1.00	0.87	0.76	0.82
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.60	0.66	0.74
	earlier estimate m, s.e. τ	1.00	1.00	1.00	1.00	0.45	0.54	0.67
Gemma 4 31B	normal, mean m, sd τ (the paper's)	1.00	0.48	0.49	0.72	0.02	0.04	0.18
	about m, give or take τ	1.00	1.00	1.00	1.00	1.01	1.22	1.45
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.90	1.03	1.15
	earlier estimate m, s.e. τ	1.00	0.94	0.94	0.96	0.09	0.23	0.41
gpt-oss-20b	normal, mean m, sd τ (the paper's)	1.00	0.64	0.52	0.52	0.23	0.29	0.34
	about m, give or take τ	1.00	0.24	0.17	0.38	0.28	0.29	0.38
	Mean: $m \pm \tau$	1.00	0.17	0.26	0.40	0.23	0.24	0.30
	earlier estimate m, s.e. τ	1.00	0.67	0.63	0.67	0.28	0.37	0.42
OLMo 2 32B	normal, mean m, sd τ (the paper's)	0.61	0.83	0.81	0.83	*0.20*	*0.29*	0.38
	about m, give or take τ	0.82	0.93	0.92	0.96	0.60	0.57	0.65
	Mean: $m \pm \tau$	0.82	0.88	0.83	0.88	0.44	0.46	0.55
	earlier estimate m, s.e. τ	0.82	0.95	0.94	0.95	0.49	0.57	0.65
Phi-4	normal, mean m, sd τ (the paper's)	0.77	0.96	0.78	0.76	*0.38*	*0.37*	*0.38*
	about m, give or take τ	0.97	1.02	1.01	1.02	0.65	0.57	0.62
	Mean: $m \pm \tau$	0.83	*0.99*	*0.87*	*0.87*	0.45	0.42	0.50
	earlier estimate m, s.e. τ	0.81	0.88	0.76	0.73	*0.47*	*0.49*	0.54
Qwen3.5 9B	normal, mean m, sd τ (the paper's)	1.00	0.81	0.76	0.80	0.26	0.27	0.44
	about m, give or take τ	1.00	0.99	1.00	0.99	0.35	0.43	0.57
	Mean: $m \pm \tau$	1.00	1.08	1.25	1.14	0.19	0.27	0.38
	earlier estimate m, s.e. τ	1.00	0.87	0.86	0.86	0.61	0.75	0.82

Specificity control. For each task the control states the value drawn for $\tau = 10$, or the one drawn for $\tau = 60$, with each of $\tau = 60$, 30, and 10, on the same $2{,}000$ tasks and demonstrations as the main run, together with its own no-description condition, at every n (Table 7). Its repeated cells reproduce the main run exactly for every model but gpt-oss-20b, whose distribution readout differs by a median of 0.8 to 1.2 (at most 39) and whose written answers differ in 12% of replies (median difference 0), likely from a newer library version. The primary contrast is the paired change in each task's prediction when the stated standard deviation goes from 60 to 10, regressed across tasks and $n = 1$, 2, and 4 on the change in the Bayes-optimal prediction, on the tasks answered in both cells: 1 for a learner that follows the stated specificity as the Bayes-optimal predictor does, 0 for one that ignores it. The Bayes-optimal reference takes the statement at face value, so with the value accurate to $\tau = 60$ but stated as $\tau = 10$ its ESS is about -0.8, and a model that ignores the stated specificity can beat it there. With the stated value fixed, its accuracy, not the stated specificity, mainly sets the ESS of the models answering without reasoning: with the value accurate to $\tau = 60$, each one's ESS is between -0.1 and 2.0 under every stated τ.

Reasoning. Both reasoning runs use the first 200 tasks of the main run and the cells of Table 8: the no-description curve at $n = 1$ to 64 (with $n = 40$ and 48 for Haiku 5.5, where the $c = 36$ ESS is read), and each description and control condition at $n = 1$, 2, and 4. gpt-oss-20b runs in vLLM with greedy decoding at low reasoning effort and up to $4{,}096$ new tokens; 99.9% of its $7{,}400$ replies reach the final channel (at least 97% in every cell), with 322 tokens on average, and the number there, read by the rule of the written readout, is its prediction. An earlier run with 384 tokens finished only 28% to 54% of the replies where a reliability is stated, and the hedge it seemed to show between a 10% and a 50% chance came from which replies finished. Claude Haiku 5.5 (`claude-haiku-5-5`) is called through the Claude Code command line on a

Table 7. **With the stated value and the demonstrations fixed, models answering without reasoning respond to the stated specificity far less than the Bayes-optimal predictor; models that reason follow it.** Specificity control, $\sigma = 60$. Response to the stated sd: the paired change in prediction from a stated standard deviation of 60 to 10, as a share of the Bayes-optimal change, over $n = 1$, 2, and 4, with 95% bootstrap intervals, for a value accurate to $\tau = 10$ or to $\tau = 60$. Right: for the value accurate to $\tau = 10$, the weight on it at $n = 1$ and the ESS (median over $n = 1$, 2, 4) under each stated τ; the Bayes-optimal predictor takes the stated τ at face value. Upper rows: 2,000 tasks, read as in Table 3; lower rows: reasoning, the first 200 tasks, on which the Bayes-optimal ESS is 2.2, 14, and 35.

	response to the stated sd		value accurate to sd 10, stated sd 60 / 30 / 10					
	value sd 10	value sd 60	weight, $n = 1$			ESS		
Bayes-optimal	1	1	0.50	0.80	0.97	2.5	13	36
gpt-oss-20b	0.02 [0.02, 0.03]	0.00 [0.00, 0.01]	0.48	0.47	0.49	2.7	2.6	2.8
Phi-4	−0.13 [−0.14, −0.12]	−0.13 [−0.13, −0.12]	0.81	0.75	0.74	12	7.2	7.6
OLMo 2 32B	0.01 [0.01, 0.01]	0.01 [0.00, 0.01]	0.70	0.70	0.70	4.8	4.6	5.3
gpt-oss-20b (written)	0.01 [−0.01, 0.04]	−0.01 [−0.03, 0.01]	0.52	0.46	0.50	2.9	3.5	3.8
Qwen3.5 9B (written)	0.12 [0.09, 0.14]	0.16 [0.13, 0.18]	0.76	0.75	0.80	3.3	3.5	4.0
Gemma 4 12B (written)	0.19 [0.16, 0.23]	0.12 [0.09, 0.14]	0.38	0.29	0.59	1.0	1.0	1.8
Gemma 4 31B (written)	0.38 [0.34, 0.41]	0.28 [0.25, 0.31]	0.43	0.49	0.72	0.2	0.3	1.4
gpt-oss-20b, low	1.03 [0.93, 1.14]	1.00 [0.96, 1.04]	0.50	0.79	0.95	0.5	8.6	> 60
Haiku 5.5, medium	0.92 [0.84, 0.99]	0.92 [0.86, 0.97]	0.36	0.69	0.74	2.1	11	28
Haiku 5.5, low	0.70 [0.59, 0.79]	0.79 [0.70, 0.86]	0.30	0.34	0.41	2.4	16	26

Table 8. **With reasoning, Haiku 5.5 comes close to the Bayes-optimal ESS; gpt-oss-20b gets the weight right but loses ESS to a few large errors.** First 200 tasks, $\sigma = 60$. Weight on the stated value at $n = 1$; ESS, median over $n = 1$, 2, and 4, with 95% bootstrap intervals, read from the model's own no-description curve, and (ref.) its ESS read from the Bayes-optimal predictor's. $\leq$, an upper bound; > 60, beyond the grid. The Bayes-optimal values are read the same way on the same tasks.

	Bayes-optimal		gpt-oss-20b, low			Haiku 5.5, medium			Haiku 5.5, low		
Description	weight	ESS	weight	ESS	ref.	weight	ESS	ref.	weight	ESS	ref.
$c = 1$	0.50	1.0	0.50	0.3 [−0.2, 1.0]	0.3 [−0.2, 1.0]	0.44	1.1 [0.6, 1.6]	1.1 [0.6, 1.6]	0.36	0.8 [0.5, 1.7]	0.7 [0.4, 1.2]
$c = 4$	0.80	3.5	0.78	1.0 [−0.1, 3.1]	1.0 [−0.1, 3.1]	0.66	3.3 [2.2, 4.7]	3.5 [2.2, 4.8]	0.34	3.4 [2.5, 5.0]	3.3 [2.1, 4.5]
$c = 36$	0.97	35	0.95	> 60 [13, >]	24 [12, 35]	0.74	28 [21, >]	28 [21, 39]	0.41	26 [22, 32]	24 [17, 31]
irrelevant	0.97	≤ -1.0	0.97	≤ -1.0	≤ -1.0	0.87	≤ -1.0	≤ -1.0	0.60	≤ -1.0	≤ -1.0
stated 10%	0.88	27	0.95	1.3 [−0.2, >]	1.2 [−0.3, 15]	0.79	33 [26, >]	33 [26, 40]	0.42	27 [22, 38]	26 [20, 37]
stated 50%	0.58	5.7	0.89	8.8 [4.2, >]	8.8 [4.2, 17]	0.55	3.8 [2.3, 6.4]	3.9 [2.4, 6.5]	0.49	5.7 [4.3, 21]	5.6 [4.2, 9.2]

subscription, one prompt per call, with no tools, no project or user instructions, and a one-line system prompt, at its default (medium) reasoning effort and at low effort; at medium it thinks before most answers (270 thinking tokens on average; 37% to 88% of replies at $n = 1$ over all conditions, and nearly all from $n = 2$), its replies are sampled at the command line's default temperature, and all but one of its 7,800 replies contain a number (three at low effort). Its no-description error is close to the Bayes-optimal predictor's through $n = 48$ (15.8 against 14.8 at $n = 16$, where two replies are far off) and rises at 64, where in 9 of 200 replies it divides the sum by 60 instead of 64. Within each cell, Haiku 5.5's replies split by whether it thinks: at one demonstration, those that think put a weight of 0.50, 0.80, and 0.97 on the relevant $c = 1$, 4, and 36 values, the Bayes-optimal weight to two decimals at either effort, and those that answer directly put 0.00 to 0.05 on it and repeat the number shown. On these three descriptions at low effort it thinks in 34% to 53% of replies, against 72% to 82% at medium, so its overall weight at one demonstration is lower and nearly flat (0.36, 0.34, 0.41); with more demonstrations it thinks more often and its weight is close to the Bayes-optimal again (0.17, 0.48, 0.86 at four, against 0.20, 0.50, 0.90). The choice to think is the model's, so the split describes its replies rather than an experiment. "Ref. ESS" reads the model's error with the description off the Bayes-optimal no-description curve instead of the model's own: the demonstrations a Bayes-optimal learner without the description would need to match it.

A frontier model inside an agent harness. The models above are open ones run locally, so that their first-token distributions can be read. The squared-error ESS needs only answers, so we also put the prompts of three relevant descriptions ($c = 1$, 4, 36; $\sigma = 60$) with $n = 1$ and 4 demonstrations, 100 tasks each, to GPT-5.6-Luna (model `gpt-5.6-luna`) through the Codex command-line agent in its read-only sandbox, on a subscription, once with its reasoning effort set to low (about 115 reasoning tokens per answer) and once with reasoning off. This measures the model as deployed in that agent: its instructions and tool definitions, about 26,000 tokens, surround the prompt, as a provider's system prompt surrounds it through an API. The model used no tool in any of these 1,200 replies, and every reply is a number alone (ten with a trailing

period; 181 with a decimal part, which we keep).

With reasoning, the weight on the stated value is the Bayes-optimal $c/(c+n)$ to within 0.03 in all six cells, the implied worth $n\,a/b$, with a and b the weights on the stated value and on the demonstrations' mean, is c to within half a demonstration (Table 9), and the error matches the Bayes-optimal predictor's to within 0.8. The prompt therefore contains what is needed to reach the Bayes-optimal weight, and this model reaches it when it reasons. With reasoning off, the same model shows the flat weight of §6: 0.72 to 0.88 at $n = 1$ and 0.40 to 0.47 at $n = 4$ whatever the stated specificity, and an implied worth of 2.6 to 5.8 demonstrations for descriptions worth 1 to 36.

To read the ESS of Definition 1 from this model's own no-description curve, we also sent the no-description prompts on the same 100 tasks at $n = 1, 2, 4, \ldots, 64$ in both modes (1,400 replies, all numbers). With reasoning, the model averaged the shown numbers with a shell command in the agent's sandbox (Python in 127 prompts, Node.js or awk in the rest) in 174 of these 700 prompts (4 at $n = 16$, 74 at $n = 32$, and 96 at $n = 64$), and its no-description error matches the Bayes-optimal predictor's within 0.1 at every n (for example 9.4 and 7.0 at $n = 32$ and 64, against 9.4 and 6.9); with reasoning off it used no tool. A more accurate no-description curve lowers the ESS read from it, so the tool use does not inflate the ESS. With reasoning, the ESS of the three descriptions at $n = 1$ and 4 matches the Bayes-optimal predictor's, read the same way from its predictions on the same tasks, to within 0.2 demonstrations for $c = 1$ and 4 and 32 against 33 for $c = 36$ (Table 10; with 100 tasks the reading itself moves the Bayes-optimal 4 to 2.4 and 2.9). With reasoning off, the $c = 36$ description is worth 5.4 demonstrations at $n = 1$ and 1.3 at $n = 4$, in the range of the open models of §6, and the $c = 1$ description hurts at $n = 4$.

Table 9. **A frontier model reaches the Bayes-optimal weight when it reasons and shows the flat weight of the open models when it does not.** GPT-5.6-Luna inside the Codex agent harness, 100 tasks per cell, relevant descriptions at $\sigma = 60$; weight on the stated value and implied worth $n\,a/b$ from the regression of §6; bootstrap standard errors of the weights are at most 0.03 with reasoning and 0.07 without.

	weight on the stated value						implied worth, demonstrations					
	$n = 1$			$n = 4$			$n = 1$			$n = 4$		
Bayes-optimal ESS c	1	4	36	1	4	36	1	4	36	1	4	36
Bayes-optimal	0.50	0.80	0.97	0.20	0.50	0.90	1	4	36	1	4	36
GPT-5.6-Luna, low reasoning	0.50	0.77	0.97	0.21	0.50	0.90	1.0	3.6	35.7	1.0	4.0	36.5
GPT-5.6-Luna, reasoning off	0.79	0.72	0.88	0.43	0.40	0.47	3.6	2.6	5.8	3.1	2.7	3.7

Table 10. **When it reasons, GPT-5.6-Luna's ESS is consistent with the Bayes-optimal ESS.** ESS at $n = 1$ and 4 demonstrations shown, read from the model's own no-description curve ($n = 1$ to 64, the same 100 tasks; with reasoning, the model averaged the numbers with a shell command in 174 of these 700 prompts, mostly at $n = 32$ and 64) as in §6, with 95% bootstrap intervals over tasks; the Bayes-optimal predictor's predictions on the same tasks are read the same way.

	$n = 1$			$n = 4$		
Bayes-optimal ESS c	1	4	36	1	4	36
Bayes-optimal, read the same way	1.0 [0.6, 1.8]	2.4 [1.6, 3.9]	33 [25, 49]	1.7 [0.5, 3.6]	2.9 [1.2, 7.7]	33 [23, 50]
GPT-5.6-Luna, low reasoning	1.1 [0.6, 1.8]	2.3 [1.4, 3.8]	32 [25, 48]	1.5 [0.3, 3.4]	3.0 [1.2, 7.7]	32 [23, 49]
GPT-5.6-Luna, reasoning off	0.1 [0.0, 0.5]	1.2 [0.8, 2.1]	5.4 [2.5, 14]	−2.2 [−2.6, −1.4]	0.1 [−0.7, 2.5]	1.3 [−0.2, 3.8]

Table 11. ESS of a description for every condition and number of demonstrations shown, for the Bayes-optimal predictor and the three models read from their distribution (2,000 tasks). "number" is the share of a model's replies that begin with a three-digit number, averaged over n; "10% wrong" and "50% wrong" mark a stated chance of being wrong. The Bayes-optimal predictor is that of §6, optimal for what the prompt states: it takes an irrelevant description with no stated chance as relevant, so its ESS can be negative. $\leq$, an upper bound; $> k$, the error with the description is below the no-description error at every n measured, up to 64, so the ESS exceeds $k = 64 - n$.

Predictor	Description	number	$n = 1$	$n = 2$	$n = 4$	$n = 8$	$n = 16$	$n = 32$	$n = 64$
Bayes-optimal	$\sigma = 60$, $c = 1$, relevant		1.2	1.3	1.3	1.3	1.2	1.5	> 0
	$\sigma = 60$, $c = 1$, irrelevant		≤ 0.0	-0.5	-0.8	-1.2	-1.7	-1.5	-1.7
	$\sigma = 60$, $c = 4$, relevant		4.5	4.4	3.9	4.6	4.6	5.2	> 0
	$\sigma = 60$, $c = 4$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	-6.2	-12	-20	-32
	$\sigma = 60$, $c = 36$, relevant		36	36	37	38	39	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	≤ -7.0	≤ -15	≤ -31	-62
	$\sigma = 30$, $c = 1$, relevant		1.0	1.2	1.4	1.6	1.4	1.5	> 0
	$\sigma = 30$, $c = 4$, relevant		4.5	4.7	5.0	5.1	4.9	5.6	> 0
	$\sigma = 30$, $c = 36$, relevant		42	43	43	43	42	> 32	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong		3.8	3.9	3.6	4.4	4.5	5.1	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong		≤ 0.0	-0.7	-1.2	-1.6	-2.2	-1.9	-2.2
	$\sigma = 60$, $c = 36$, relevant, 10% wrong		13	22	26	35	38	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong		≤ 0.0	-0.9	-1.7	-2.8	-5.4	-6.9	-10
	$\sigma = 60$, $c = 4$, relevant, 50% wrong		1.9	2.2	2.9	3.7	4.1	4.5	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong		≤ 0.0	-0.2	-0.4	-0.5	-0.8	-0.6	-0.7
	$\sigma = 60$, $c = 36$, relevant, 50% wrong		3.1	5.8	12	21	31	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong		≤ 0.0	-0.3	-0.7	-1.2	-2.4	-2.8	-4.5
gpt-oss-20b	$\sigma = 60$, $c = 1$, relevant	0.98	1.0	0.9	-0.1	-1.5	-9.2	-25	-57
	$\sigma = 60$, $c = 1$, irrelevant	0.98	≤ 0.0	-0.3	-1.5	-3.6	-10	-25	-59
	$\sigma = 60$, $c = 4$, relevant	0.98	2.3	1.7	2.0	2.9	-7.5	-24	-48
	$\sigma = 60$, $c = 4$, irrelevant	0.97	≤ 0.0	-0.3	-1.4	-3.7	-11	-26	-59
	$\sigma = 60$, $c = 36$, relevant	0.98	2.8	2.3	3.8	17	3.7	-12	-17
	$\sigma = 60$, $c = 36$, irrelevant	0.98	≤ 0.0	-0.5	-1.5	-3.7	-11	-26	-60
	$\sigma = 30$, $c = 1$, relevant	0.99	0.8	0.4	-0.3	-1.8	-9.9	-26	-59
	$\sigma = 30$, $c = 4$, relevant	0.98	1.2	1.3	1.9	3.4	-8.1	-24	-44
	$\sigma = 30$, $c = 36$, relevant	0.99	1.6	2.3	4.7	31	3.0	-13	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.96	2.2	1.8	0.0	-0.2	-8.1	-24	-52
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.6	-1.1	-2.2	-9.6	-25	-58
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.97	2.4	2.4	0.5	2.8	-3.6	-20	-33
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.7	-1.1	-2.3	-9.7	-25	-59
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.95	2.3	1.7	0.1	-0.5	-8.3	-24	-53
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	0.94	≤ 0.0	-0.5	-1.1	-2.3	-9.7	-25	-58
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.95	2.8	2.3	0.7	1.2	-4.5	-21	-37
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.95	≤ 0.0	-0.6	-1.1	-2.3	-9.7	-25	-59
Phi-4	$\sigma = 60$, $c = 1$, relevant	0.87	0.8	0.7	-0.2	-0.4	-1.0	-3.1	-35
	$\sigma = 60$, $c = 1$, irrelevant	0.86	≤ 0.0	≤ -1.0	≤ -3.0	-4.9	-6.1	-8.8	-40
	$\sigma = 60$, $c = 4$, relevant	0.85	5.8	4.1	2.4	2.1	0.7	-4.6	-33
	$\sigma = 60$, $c = 4$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.7	-6.2	-13	-42
	$\sigma = 60$, $c = 36$, relevant	0.86	15	7.6	4.7	5.6	8.5	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.5	-6.4	-12	-42
	$\sigma = 30$, $c = 1$, relevant	0.87	1.1	0.8	-0.3	-0.3	-1.0	-1.8	-15
	$\sigma = 30$, $c = 4$, relevant	0.86	4.8	3.5	2.0	1.6	0.6	-1.9	-2.2
	$\sigma = 30$, $c = 36$, relevant	0.88	9.7	6.0	3.8	4.4	8.4	17	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.87	6.0	5.5	5.3	8.3	15	> 32	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	0.85	≤ 0.0	≤ -1.0	-2.1	-1.2	2.9	-3.2	> 0
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.87	10	8.7	8.2	17	> 48	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	0.86	≤ 0.0	≤ -1.0	-2.4	-1.7	1.3	-4.7	> 0
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.87	6.0	6.4	7.7	17	> 48	> 32	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	0.85	≤ 0.0	≤ -1.0	-1.3	1.5	16	> 32	> 0
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.88	9.2	9.7	11	> 56	> 48	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.86	≤ 0.0	≤ -1.0	-1.8	0.6	13	> 32	> 0
OLMo 2 32B	$\sigma = 60$, $c = 1$, relevant	0.90	1.9	1.8	2.0	2.3	5.8	> 32	> 0
	$\sigma = 60$, $c = 1$, irrelevant	0.89	≤ 0.0	-0.5	-0.2	-0.2	1.7	25	> 0
	$\sigma = 60$, $c = 4$, relevant	0.91	5.8	3.4	3.4	4.5	9.3	> 32	> 0
	$\sigma = 60$, $c = 4$, irrelevant	0.90	≤ 0.0	-0.7	-0.3	-0.5	0.3	17	> 0
	$\sigma = 60$, $c = 36$, relevant	0.91	11	4.6	5.3	6.8	15	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant	0.90	≤ 0.0	≤ -1.0	-0.6	-0.5	-0.2	15	> 0
	$\sigma = 30$, $c = 1$, relevant	0.92	1.8	1.9	1.7	1.7	4.4	24	> 0
	$\sigma = 30$, $c = 4$, relevant	0.92	4.8	3.3	3.1	4.0	9.5	> 32	> 0
	$\sigma = 30$, $c = 36$, relevant	0.93	7.0	4.4	4.0	5.9	14	> 32	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.96	6.0	4.6	3.8	4.5	5.8	23	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.9	0.0	0.0	-0.4	4.1	> 0
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.96	12	5.9	6.0	6.3	8.9	31	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	0.96	≤ 0.0	≤ -1.0	-0.3	-0.1	-0.8	0.3	> 0
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.93	5.3	4.2	3.5	4.0	5.4	27	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	0.92	≤ 0.0	-0.5	0.1	0.1	0.0	11	> 0
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.93	7.8	5.4	4.9	5.6	7.9	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.92	≤ 0.0	-0.8	-0.1	0.0	-0.3	7.8	> 0

Table 12. ESS of a description for every condition and number of demonstrations shown, for the three models read from the number they write (2,000 tasks; a reply without a number is left out); columns and the Bayes-optimal values as in Table 11.

Predictor	Description	number	$n=1$	$n=2$	$n=4$	$n=8$	$n=16$	$n=32$	$n=64$
Qwen3.5 9B (written)	$\sigma=60, c=1$, relevant	1.00	0.2	0.3	-0.9	-2.8	-9.3	-24	-51
	$\sigma=60, c=1$, irrelevant	1.00	≤ 0.0	≤ -1.0	-2.7	-4.8	-9.7	-23	-49
	$\sigma=60, c=4$, relevant	1.00	3.9	2.7	1.3	-0.7	-6.2	-18	-44
	$\sigma=60, c=4$, irrelevant	1.00	≤ 0.0	≤ -1.0	-1.7	-3.8	-9.4	-24	-49
	$\sigma=60, c=36$, relevant	1.00	6.5	4.0	2.8	6.9	17	>32	>0
	$\sigma=60, c=36$, irrelevant	1.00	≤ 0.0	≤ -1.0	-2.8	-5.1	-10	-24	-51
	$\sigma=30, c=1$, relevant	1.00	≤ 0.0	0.2	-0.7	-2.5	-8.7	-22	-49
	$\sigma=30, c=4$, relevant	1.00	3.4	2.7	1.5	-0.3	-4.5	-14	-40
	$\sigma=30, c=36$, relevant	1.00	4.7	4.3	3.0	9.2	14	>32	>0
	$\sigma=60, c=4$, relevant, 10% wrong	1.00	3.8	-0.4	-0.7	-1.8	-8.5	-21	-47
	$\sigma=60, c=4$, irrelevant, 10% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	-4.9	-10	-24	-50
	$\sigma=60, c=36$, relevant, 10% wrong	1.00	5.3	0.4	0.6	-0.3	-3.3	-4.2	-6.9
	$\sigma=60, c=36$, irrelevant, 10% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	-5.2	-10	-25	-53
	$\sigma=60, c=4$, relevant, 50% wrong	1.00	4.5	-0.1	-0.4	-1.9	-9.0	-23	-50
	$\sigma=60, c=4$, irrelevant, 50% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	-6.2	-12	-26	-54
	$\sigma=60, c=36$, relevant, 50% wrong	1.00	7.3	0.8	1.0	-0.1	-4.0	-4.2	-12
	$\sigma=60, c=36$, irrelevant, 50% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	-7.0	-13	-27	-56
Gemma 4 12B (written)	$\sigma=60, c=1$, relevant	1.00	0.0	0.0	-0.1	-0.6	-3.2	-5.4	-21
	$\sigma=60, c=1$, irrelevant	1.00	≤ 0.0	-0.4	-1.3	-1.3	-4.4	-6.8	-24
	$\sigma=60, c=4$, relevant	1.00	0.3	0.5	1.4	1.1	-0.6	-2.7	-13
	$\sigma=60, c=4$, irrelevant	1.00	≤ 0.0	-0.4	-1.1	-1.2	-4.1	-6.9	-23
	$\sigma=60, c=36$, relevant	1.00	1.8	1.2	2.7	3.6	3.6	4.9	>0
	$\sigma=60, c=36$, irrelevant	1.00	≤ 0.0	-0.8	-1.3	-1.3	-4.9	-7.9	-26
	$\sigma=30, c=1$, relevant	1.00	≤ 0.0	-0.2	-0.1	-0.5	-3.2	-7.9	-27
	$\sigma=30, c=4$, relevant	1.00	0.7	0.8	2.1	1.5	-0.2	-4.6	-18
	$\sigma=30, c=36$, relevant	1.00	4.0	1.7	3.4	3.7	2.4	-1.8	-8.9
	$\sigma=60, c=4$, relevant, 10% wrong	1.00	0.1	0.7	1.4	0.4	-1.0	-3.1	-7.3
	$\sigma=60, c=4$, irrelevant, 10% wrong	1.00	≤ 0.0	-0.2	-0.4	-0.5	-2.2	-5.4	-13
	$\sigma=60, c=36$, relevant, 10% wrong	1.00	0.6	1.3	2.3	1.6	0.7	-0.6	-1.3
	$\sigma=60, c=36$, irrelevant, 10% wrong	1.00	≤ 0.0	-0.2	-0.5	-0.5	-2.5	-5.8	-15
	$\sigma=60, c=4$, relevant, 50% wrong	1.00	1.0	0.6	1.7	1.0	-0.1	-1.7	-4.5
	$\sigma=60, c=4$, irrelevant, 50% wrong	1.00	≤ 0.0	-0.3	-0.4	-0.4	-1.3	-4.1	-9.6
	$\sigma=60, c=36$, relevant, 50% wrong	1.00	5.1	1.0	2.7	2.3	2.4	1.1	>0
	$\sigma=60, c=36$, irrelevant, 50% wrong	1.00	≤ 0.0	-0.3	-0.6	-0.4	-1.9	-4.8	-13
Gemma 4 31B (written)	$\sigma=60, c=1$, relevant	1.00	0.1	0.0	-0.1	-0.5	-1.4	-5.1	-13
	$\sigma=60, c=1$, irrelevant	1.00	≤ 0.0	-0.1	-0.1	-0.5	-1.0	-4.5	-11
	$\sigma=60, c=4$, relevant	1.00	0.8	0.1	0.2	-0.5	-1.0	-4.3	-12
	$\sigma=60, c=4$, irrelevant	1.00	≤ 0.0	-0.2	-0.2	-0.4	-1.0	-4.4	-10
	$\sigma=60, c=36$, relevant	1.00	2.6	0.6	1.4	0.9	2.9	3.4	-2.9
	$\sigma=60, c=36$, irrelevant	1.00	≤ 0.0	≤ -1.0	-0.8	-0.6	-1.1	-2.7	-4.8
	$\sigma=30, c=1$, relevant	1.00	≤ 0.0	0.0	-0.1	-0.7	-1.2	-3.0	-8.1
	$\sigma=30, c=4$, relevant	1.00	0.5	0.0	0.1	-0.5	-1.3	-3.4	-8.3
	$\sigma=30, c=36$, relevant	1.00	4.3	1.5	2.1	2.3	3.3	3.3	-0.4
	$\sigma=60, c=4$, relevant, 10% wrong	1.00	0.1	0.0	0.0	-0.7	-1.9	-5.0	-15
	$\sigma=60, c=4$, irrelevant, 10% wrong	1.00	≤ 0.0	0.0	-0.1	-0.4	-1.2	-4.1	-9.5
	$\sigma=60, c=36$, relevant, 10% wrong	1.00	1.0	0.0	0.6	-0.3	-1.4	-4.8	-12
	$\sigma=60, c=36$, irrelevant, 10% wrong	1.00	≤ 0.0	0.0	-0.2	-0.4	-0.9	-3.0	-6.2
	$\sigma=60, c=4$, relevant, 50% wrong	1.00	0.2	0.0	0.0	-0.7	-1.7	-4.6	-17
	$\sigma=60, c=4$, irrelevant, 50% wrong	1.00	≤ 0.0	0.0	-0.2	-0.4	-1.2	-3.6	-12
	$\sigma=60, c=36$, relevant, 50% wrong	1.00	1.0	0.0	0.5	-0.3	-1.5	-4.5	-15
	$\sigma=60, c=36$, irrelevant, 50% wrong	1.00	≤ 0.0	0.0	-0.1	-0.4	-1.0	-2.9	-8.5

Table 13. Weight on the stated value for every condition and number of demonstrations shown, for the Bayes-optimal predictor and the three models read from their distribution; "number" as in Table 11. A cell is in italics where the model begins fewer than three quarters of its replies with a number at that n.

Predictor	Description	number	$n=0$	$n=1$	$n=2$	$n=4$	$n=8$	$n=16$	$n=32$	$n=64$
Bayes-optimal	$\sigma=60, c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60, c=1$, irrelevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60, c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60, c=4$, irrelevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60, c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60, c=36$, irrelevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=30, c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=30, c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=30, c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60, c=4$, relevant, 10% wrong		0.90	0.71	0.59	0.45	0.30	0.19	0.10	0.05
	$\sigma=60, c=4$, irrelevant, 10% wrong		0.90	0.27	0.14	0.08	0.04	0.02	0.01	0.00
	$\sigma=60, c=36$, relevant, 10% wrong		0.90	0.84	0.84	0.82	0.77	0.65	0.51	0.34
	$\sigma=60, c=36$, irrelevant, 10% wrong		0.90	0.26	0.14	0.07	0.03	0.01	0.00	0.00
	$\sigma=60, c=4$, relevant, 50% wrong		0.50	0.46	0.39	0.32	0.22	0.14	0.08	0.04
	$\sigma=60, c=4$, irrelevant, 50% wrong		0.50	0.12	0.07	0.03	0.02	0.01	0.00	0.00
	$\sigma=60, c=36$, relevant, 50% wrong		0.50	0.55	0.58	0.60	0.59	0.53	0.42	0.29
	$\sigma=60, c=36$, irrelevant, 50% wrong		0.50	0.12	0.06	0.03	0.01	0.00	0.00	0.00
gpt-oss-20b	$\sigma=60, c=1$, relevant	0.98	0.94	0.53	0.29	0.23	0.16	0.12	0.12	0.23
	$\sigma=60, c=1$, irrelevant	0.98	0.93	0.59	0.25	0.18	0.09	0.05	0.04	0.10
	$\sigma=60, c=4$, relevant	0.98	0.92	0.48	0.32	0.26	0.21	0.19	0.18	0.31
	$\sigma=60, c=4$, irrelevant	0.97	0.92	0.57	0.25	0.16	0.08	0.04	0.04	0.09
	$\sigma=60, c=36$, relevant	0.98	0.94	0.49	0.35	0.30	0.25	0.23	0.22	0.35
	$\sigma=60, c=36$, irrelevant	0.98	0.95	0.60	0.25	0.15	0.07	0.04	0.04	0.09
	$\sigma=30, c=1$, relevant	0.99	0.98	0.43	0.23	0.17	0.12	0.09	0.11	0.21
	$\sigma=30, c=4$, relevant	0.98	0.98	0.38	0.28	0.23	0.19	0.18	0.19	0.31
	$\sigma=30, c=36$, relevant	0.99	0.99	0.37	0.32	0.27	0.23	0.22	0.24	0.37
	$\sigma=60, c=4$, relevant, 10% wrong	0.96	0.91	0.45	0.33	0.15	0.13	0.12	0.13	0.22
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.96	0.91	0.55	0.34	0.11	0.05	0.02	0.02	0.04
	$\sigma=60, c=36$, relevant, 10% wrong	0.97	0.92	0.44	0.34	0.17	0.15	0.15	0.16	0.25
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.96	0.92	0.53	0.34	0.10	0.04	0.02	0.02	0.04
	$\sigma=60, c=4$, relevant, 50% wrong	0.95	0.79	0.46	0.32	0.16	0.12	0.12	0.12	0.20
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.94	0.79	0.53	0.32	0.11	0.04	0.02	0.01	0.04
	$\sigma=60, c=36$, relevant, 50% wrong	0.95	0.83	0.46	0.32	0.17	0.14	0.15	0.15	0.23
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.95	0.83	0.53	0.31	0.11	0.04	0.02	0.02	0.04
Phi-4	$\sigma=60, c=1$, relevant	0.87	*0.97*	0.82	0.62	0.43	0.28	0.18	0.13	0.12
	$\sigma=60, c=1$, irrelevant	0.86	*0.97*	0.83	0.64	0.42	0.22	0.11	0.06	0.05
	$\sigma=60, c=4$, relevant	0.85	*0.98*	0.76	0.55	0.37	0.26	0.19	0.15	0.16
	$\sigma=60, c=4$, irrelevant	0.84	*0.98*	0.79	0.56	0.36	0.18	0.09	0.06	0.05
	$\sigma=60, c=36$, relevant	0.86	*0.99*	0.74	0.53	0.38	0.29	0.23	0.20	0.20
	$\sigma=60, c=36$, irrelevant	0.84	*0.99*	0.78	0.53	0.33	0.16	0.09	0.06	0.04
	$\sigma=30, c=1$, relevant	0.87	*0.99*	0.77	0.58	0.38	0.22	0.13	0.09	0.09
	$\sigma=30, c=4$, relevant	0.86	*1.00*	0.69	0.51	0.34	0.23	0.16	0.13	0.13
	$\sigma=30, c=36$, relevant	0.88	*1.00*	0.65	0.48	0.32	0.23	0.18	0.17	0.16
	$\sigma=60, c=4$, relevant, 10% wrong	0.87	*0.94*	0.62	0.46	0.30	0.21	0.15	0.13	0.11
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.85	*0.95*	0.62	0.44	0.26	0.13	0.06	0.04	0.03
	$\sigma=60, c=36$, relevant, 10% wrong	0.87	*0.95*	0.61	0.46	0.33	0.25	0.20	0.18	0.16
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.86	*0.96*	0.61	0.43	0.26	0.13	0.06	0.04	0.03
	$\sigma=60, c=4$, relevant, 50% wrong	0.87	*0.92*	0.57	0.43	0.29	0.20	0.15	0.13	0.11
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.85	*0.92*	0.57	0.42	0.24	0.12	0.06	0.04	0.03
	$\sigma=60, c=36$, relevant, 50% wrong	0.88	*0.93*	0.56	0.44	0.33	0.24	0.21	0.19	0.17
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.86	*0.94*	0.57	0.41	0.24	0.12	0.06	0.04	0.03
OLMo 2 32B	$\sigma=60, c=1$, relevant	0.90	*0.99*	0.72	0.33	0.25	0.16	0.09	0.05	0.05
	$\sigma=60, c=1$, irrelevant	0.89	*0.99*	0.77	0.29	0.16	0.09	0.05	0.02	0.02
	$\sigma=60, c=4$, relevant	0.91	*1.00*	0.70	0.37	0.30	0.21	0.14	0.08	0.07
	$\sigma=60, c=4$, irrelevant	0.90	*1.00*	0.75	0.29	0.15	0.08	0.04	0.02	0.01
	$\sigma=60, c=36$, relevant	0.91	*1.00*	0.70	0.40	0.34	0.26	0.19	0.12	0.10
	$\sigma=60, c=36$, irrelevant	0.90	*1.00*	0.76	0.29	0.17	0.08	0.04	0.02	0.02
	$\sigma=30, c=1$, relevant	0.92	*0.99*	0.69	0.32	0.23	0.14	0.09	0.04	0.04
	$\sigma=30, c=4$, relevant	0.92	*1.00*	0.65	0.34	0.27	0.19	0.13	0.06	0.06
	$\sigma=30, c=36$, relevant	0.93	*1.00*	0.64	0.36	0.32	0.23	0.16	0.08	0.09
	$\sigma=60, c=4$, relevant, 10% wrong	0.96	1.01	0.72	0.42	0.29	0.18	0.11	0.09	0.05
	$\sigma=60, c=4$, irrelevant, 10% wrong	0.96	1.01	0.81	0.32	0.14	0.06	0.03	0.02	0.01
	$\sigma=60, c=36$, relevant, 10% wrong	0.96	1.01	0.71	0.44	0.33	0.23	0.15	0.13	0.07
	$\sigma=60, c=36$, irrelevant, 10% wrong	0.96	1.01	0.82	0.33	0.15	0.07	0.03	0.02	0.01
	$\sigma=60, c=4$, relevant, 50% wrong	0.93	1.02	0.65	0.39	0.27	0.16	0.09	0.07	0.04
	$\sigma=60, c=4$, irrelevant, 50% wrong	0.92	1.01	0.71	0.29	0.13	0.06	0.02	0.01	0.01
	$\sigma=60, c=36$, relevant, 50% wrong	0.93	1.02	0.64	0.42	0.31	0.21	0.13	0.11	0.06
	$\sigma=60, c=36$, irrelevant, 50% wrong	0.92	1.02	0.72	0.30	0.14	0.06	0.03	0.02	0.01